



# Fréchet and Gateaux $gH$ -differentiability for interval valued functions of multiple variables

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## ABSTRACT

In this work we extend concepts of differentiability for Interval-Valued functions (IV-functions) of multiple variables, based on generalized Hukuhara gH-difference. In this context, we introduce a new concept of gH-linearity and characterize the class of gH-linear IV-functions, which are fundamental for a general approach to Fréchet-type and Gateaux-type gH-differentiability of the first order. We moreover consider vector IV-functions and outline the definition of the gH-Jacobian; by representing intervals and IV-functions in midpoint-radius notation, we establish properties and relations between pointwise, Fréchet and Gateaux gH-differentiability. Finally, higher-order differentiability and gH-Hessian matrix are considered. These intuitive concepts are mathematically and computationally easy to work with.

## 1. Introduction

In interval arithmetic, Moore's development is usually accepted as the standard interval arithmetic (SIA); a complete treatment of SIA can be found in Moore [29] and, e.g., in Alefeld and Herzberger [2] books. Several other approaches to interval arithmetic have been proposed in the old and recent literature.

The difference between two intervals has been a key challenge in interval arithmetic. The Hukuhara difference (H-difference) has been proposed in 1967, [22]; a definition of extended arithmetic (also called non-standard interval arithmetic) was suggested in 1977 by S. Markov [26], with application to the calculus for interval-valued functions in 1979 [27]. The extension of H-difference to the fuzzy case was developed by Puri and Ralescu in 1983 [32], adopted in the book by Diamond and Kloeden in 1990 [15] and used by Bede and Gal in 2005 [8] to define generalized fuzzy differentiability (GH-differentiability, note the big G in GH-); the fuzzy gH-difference, in the form used here, was introduced in [41], [43] and developed further in [42] and [9], [44], where the  $gH$ -differentiability (note the small g in  $gH$ -) for fuzzy-valued functions of a single variable is introduced (see also [10], [46] on multidimensional set-valued functions, not considered here). More recently, the difference operation for intervals (in the setting of constraint interval arithmetic - CIA) has been revisited in [13] and [28], following ideas from [23] (see also [24] for further references). It is to be remarked that the proposed interval differences, interpreted as binary operations, all give the same resulting interval.

The concepts of differentiability for fuzzy-valued functions of a single variable usually have their origin on the corresponding definitions for interval-valued functions, as a first stage, as the reader can check from the above references on the topic (see, in

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particular, [8], [9], [11], [14]). In a similar way, the differentiability of interval and fuzzy-valued functions in the multidimensional case, based on the  $gH$ -differences, has been analyzed extended from the single variable case; to this matter, Chalco-Cano et al. [12], Luhandjula and Rangoaga [25], Ahmad et al. [1], [3], among others, proposed concepts of  $gH$ -differentiability for interval-valued functions of several variables, based on the partial  $gH$ -derivatives and on the existence of the gradient. In 2019, Stefanini and Arana [45] introduced the  $gH$ -differentiability for interval and fuzzy-valued functions of several variables, by means of the  $gH$ -differential, that unifies and extends the previous ones; the corresponding existence of  $gH$ -gradient and partial  $gH$ -derivatives are deduced from that, as expected. Recently, Roy et al. [36], based on a work by Roy and Panda [35], proposed a definition of  $gH$ -differentiability in several variables based on the existence of a gradient and a Minkowski-type combination of the partial  $gH$ -derivatives (the components of the gradient), which they improperly call a 'linear' function; however, their definition is more restrictive than the one given by Stefanini and Arana [45] on  $gH$ -differentiability for interval-valued functions of several variables, and presents some existence issues, as will be shown later.

The aim of this work is the following. Based on the previous comments on differentiability for  $IV$ -functions in several variables, we propose a new concept of  $gH$ -linearity, as well as a characterization of the class of  $gH$ -linear  $IV$ -functions, with the corresponding discussion with other related concepts in the current literature. Thanks to the latter, we offer a general approach to Fréchet-type and Gateaux-type  $gH$ -differentiability of the first order, with properties and relations between them. Furthermore, we extend these concepts to vector  $IV$ -functions, and provide the definition of the  $gH$ -Jacobian. To this matter, we propose the following structure for this work. In Section 2, we provide some preliminaries on arithmetic and partial orders on intervals. In Section 3, we introduce a new concept of linearity in the space on interval-valued functions, called  $gH$ -linearity, based on equivalence classes of a vector space. Thanks to that, in Section 4 (the main part of paper), we define and justify the Fréchet and Gateaux type differentiability for an  $IV$ -function in several variables, with a study of their properties in depth; our study continues with the analysis of relationships among partial derivatives, gradient,  $gH$ -tangency, convexity, and gives some illustrative examples. In Section 5, we outline the extension of  $gH$ -differentiability to vector valued and to second order  $gH$ -Hessian matrix for  $IV$ - functions, with an example. A brief Section 6 concludes the work and short Appendix A describes the notation we adopt for Intervals throughout the text.

## 2. Preliminaries on intervals

We denote by  $I(\mathbb{R})$  the set of compact intervals in  $\mathbb{R}$ , i.e., in the usual end-points representation,

$$I(\mathbb{R}) = \{ [\underline{a}, \bar{a}] : \underline{a}, \bar{a} \in \mathbb{R} \text{ and } \underline{a} \leq \bar{a} \}. \quad (1)$$

For real intervals we will extensively adopt the well-known *midpoint-radius* (or, simply, *midpoint*) representation: for a given interval  $A = [\underline{a}, \bar{a}]$ , denote the midpoint  $\hat{a}$  and the radius  $\tilde{a}$ , respectively, by  $\hat{a} = \frac{\underline{a} + \bar{a}}{2}$  and  $\tilde{a} = \frac{\bar{a} - \underline{a}}{2}$  so that  $\underline{a} = \hat{a} - \tilde{a}$  and  $\bar{a} = \hat{a} + \tilde{a}$ ; correspondingly, we will denote an interval by  $A = [\underline{a}, \bar{a}]$  or, in midpoint notation, by  $A = (\hat{a}; \tilde{a})$  so that we will also write

$$I(\mathbb{R}) = \{ (\hat{a}; \tilde{a}) : \hat{a}, \tilde{a} \in \mathbb{R} \text{ and } \tilde{a} \geq 0 \}. \quad (2)$$

It is elementary to prove that the two representations are equivalent.

**Proposition 1.** The linear transformation  $[\underline{x}, \bar{x}] \longrightarrow (\hat{x}; \tilde{x})$  defined by

$$\begin{cases} \hat{x} = (\bar{x} + \underline{x}) / 2 \\ \tilde{x} = (\bar{x} - \underline{x}) / 2 \end{cases} \iff \begin{cases} \underline{x} = \hat{x} - \tilde{x} \\ \bar{x} = \hat{x} + \tilde{x} \end{cases} \quad (3)$$

is a bijection such that  $\underline{x} \leq \bar{x} \iff \tilde{x} \geq 0$ ; it is represented by the invertible matrix  $\mathbf{M} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$  with inverse  $\mathbf{M}^{-1} = 2\mathbf{M}^T$  and satisfies  $\mathbf{M}^T \mathbf{M} = \mathbf{M} \mathbf{M}^T = \frac{1}{2} \mathbf{I}$ .

**Remark 2.** The existing literature in Interval Analysis and related areas of research tends to adopt different symbols to denote intervals, sets of intervals, arithmetic operations and related space structures, etc. We face the necessity for a flexible but (relatively) complete set of symbols for jointly use of the two end-point and mid-point representations, to possibly allow for non-confusing but rapid typing and avoid, e.g.,  $\text{rad}(A)$ ,  $\text{width}(A)$ ,  $\text{len}(A)$  and similar to denote important structure-terms of intervals, operations, equations and writing formulae.

Please, refer to Appendix A, where we describe our notational choice for this paper.

Using midpoint notation, the Minkowski addition and scalar multiplication are the following, given  $A = [\underline{a}, \bar{a}] = (\hat{a}; \tilde{a})$ ,  $B = [\underline{b}, \bar{b}] = (\hat{b}; \tilde{b})$  and  $\tau \in \mathbb{R}$ :

$$A \oplus_M B = (\hat{a} + \hat{b}; \tilde{a} + \tilde{b}), \tau A = \{\tau a : a \in A\} = (\tau \hat{a}; |\tau| \tilde{a}), \quad (4)$$

$$-A = (-\hat{a}; \tilde{a}), A \ominus_M B = A \oplus_M (-B) = (\hat{a} - \hat{b}; \tilde{a} + \tilde{b}). \quad (5)$$

The  $gH$ -difference of two intervals  $A$  and  $B$  is the following:

$$A \ominus_{gH} B = C \iff \begin{cases} (a) A = B \oplus_M C, \\ \text{or } (b) B = A \oplus_M (-1)C. \end{cases}$$

The  $gH$ -difference of two intervals always exists and is given by

$$\begin{aligned} A \ominus_{gH} B &= [\min\{\underline{a} - \underline{b}, \bar{a} - \bar{b}\}, \max\{\underline{a} - \underline{b}, \bar{a} - \bar{b}\}] \\ &= (\hat{a} - \hat{b}; |\tilde{a} - \tilde{b}|). \end{aligned}$$

We have  $A \ominus_{gH} B \subseteq A \ominus_M B$ . It is immediate that the  $gH$ -difference of an interval and itself is zero, that is,  $A \ominus_{gH} A = [0, 0]$ .

**Remark 3.** In this paper, the  $gH$ -type operations are not intended to give alternative arithmetic for intervals; instead, as we will see, they are introduced and used in conjunction with Minkowski (or SIA) setting.

Given  $A, B \in I(\mathbb{R})$ , the Pompeiu-Hausdorff distance is defined by

$$H(A, B) = \max \left[ \max_{a \in A} d(a, B), \max_{b \in B} d(b, A) \right] \quad (6)$$

with  $d(a, B) = \min_{b \in B} |a - b|$ . It is known (see [42]) that  $H(A, B) = \|A \ominus_{gH} B\|_H$  where, for  $C \in I(\mathbb{R})$ , the quantity  $\|C\|_H = \max\{|c|; c \in C\}$  is called the *magnitude* of  $C$ . We have the following properties:

$$H(A, B) = 0 \iff A \ominus_{gH} B = 0 \iff A = B$$

$$H(\tau A, \tau B) = |\tau| H(A, B), \forall \tau \in \mathbb{R},$$

$$H(A \oplus_M C, B \oplus_M C) = H(A, B),$$

$$H(A \oplus_M B, C \oplus_M D) \leq H(A, C) + H(B, D).$$

We will use the following  $gH$ -addition (it results to be equal to the Markov's 1977 extended addition), defined as the interval  $C$  such that  $A \oplus_{gH} B = C \iff A \ominus_{gH} (-B) = C$  and

$$\begin{aligned} A \oplus_{gH} B &= [\min\{\underline{a} + \underline{b}, \bar{a} + \bar{b}\}, \max\{\underline{a} + \underline{b}, \bar{a} + \bar{b}\}] \\ &= (\hat{a} + \hat{b}; |\tilde{a} - \tilde{b}|) \subset A \oplus_M B. \end{aligned}$$

Obviously,  $A \oplus_{gH} B \subseteq A \oplus_M B$  and  $A \oplus_M A = 2A$  while  $A \oplus_{gH} A = (2\hat{a}; 0)$ ; we could see that Minkowski addition is reproductive, while  $gH$ -addition is not. Furthermore,  $\oplus_{gH}$  is commutative but not associative; this will be important when considering its iterated application.

The set  $I(\mathbb{R})$  of intervals can be ordered by several partial orders (see e.g., [49]). The two basic orders, generally adopted in the analysis of interval-valued functions, define comparability of two intervals either by considering their relative position on the real line (by extending the ordinary order of real numbers), or by taking the ordinary set inclusion.

**Definition 4.** Given  $A = [\underline{a}, \bar{a}]$ ,  $B = [\underline{b}, \bar{b}] \in I(\mathbb{R})$ , we say that

- (i) ( $\leq_{LU}$ -lattice order):  $A \leq_{LU} B$  if and only if  $\underline{a} \leq \underline{b}$  and  $\bar{a} \leq \bar{b}$ ,
- (ii) ( $\leq_{LU}$ -strict order):  $A \leq_{LU} B$  if and only if  $A \leq_{LU} B$  and  $A \neq B$ ,
- (iii) ( $<_{LU}$ -strong order):  $A <_{LU} B$  if and only if  $\underline{a} < \underline{b}$  and  $\bar{a} < \bar{b}$ .

The order  $A < B \iff \bar{a} < \underline{b}$ , is considered as a special case of (iii).

In terms of set inclusion, we have that

- (a) ( $\subseteq$ -lattice order):  $A \subseteq B$ , if and only if  $\underline{a} \geq \underline{b}$  and  $\bar{a} \leq \bar{b}$ ;
- (b) ( $\subseteq$ -strict order):  $A \subseteq B$ , if and only if (a) is satisfied with  $A \neq B$ ;
- (c) ( $\subset$ -strong order):  $A \subset B$ , if and only if  $\underline{a} > \underline{b}$  and  $\bar{a} < \bar{b}$ .

With respect to the LU partial order, it is well known that

$$A \leq_{LU} B \iff A \ominus_{gH} B \leq_{LU} 0.$$

Using midpoint notation, we have the following expressions for the partial orders above (see [19], [45]):

**Proposition 5.** Let  $A, B \in I(\mathbb{R})$  denoted by  $A = (\hat{a}; \tilde{a})$ ,  $B = (\hat{b}; \tilde{b})$ ; then

- $A \leq_{LU} B$  if and only if
  - (i)  $\begin{cases} \tilde{b} \leq \tilde{a} + (\hat{b} - \hat{a}); \\ \tilde{b} \geq \tilde{a} - (\hat{b} - \hat{a}); \end{cases}$
- $A \leq_{LU} B$  if and only if (i) is satisfied with at least one strict inequality;
- $A <_{LU} B$  if and only if
  - (iii)  $\begin{cases} \tilde{b} < \tilde{a} + (\hat{b} - \hat{a}); \\ \tilde{b} > \tilde{a} - (\hat{b} - \hat{a}); \end{cases}$
- $A \subseteq B$ , if and only if
  - (a)  $\begin{cases} \tilde{b} \geq \tilde{a} + (\hat{b} - \hat{a}); \\ \tilde{b} \geq \tilde{a} - (\hat{b} - \hat{a}); \end{cases}$
- $A \subseteq B$ , if and only if (a) are satisfied with at least one strict inequality (i.e., with  $A \neq B$ );
- $A \subset B$ , if and only if
  - (c)  $\begin{cases} \tilde{b} > \tilde{a} + (\hat{b} - \hat{a}); \\ \tilde{b} > \tilde{a} - (\hat{b} - \hat{a}). \end{cases}$

**Proof.** We have to express the conditions of Definition 4 in terms of  $\underline{a} = \hat{a} - \tilde{a}$  and  $\bar{a} = \hat{a} + \tilde{a}$  (and similarly for B); e.g., to prove (a) for  $A \subseteq B$ , we have  $(\underline{a} \geq \underline{b}, \bar{a} \leq \bar{b})$  if and only if  $(\hat{a} - \tilde{a} \geq \hat{b} - \tilde{b}, \hat{a} + \tilde{a} \leq \hat{b} + \tilde{b})$ , obtaining, after moving terms from left to right,  $(\tilde{b} \geq \tilde{a} + (\hat{b} - \hat{a}), \tilde{b} \geq \tilde{a} - (\hat{b} - \hat{a}))$ .  $\square$

For simplicity, we frequently omit the subscript  $()_{LU}$  and we refer to the LU-orders simply by notations  $\leq$ , or  $\leq$ , or  $<$ .

In particular, if  $\hat{a} = \hat{b}$ , we can have  $A \leq B$  only if  $A = B$ , it is not possible to have  $A \leq A$  nor  $A < A$  and condition  $A \subseteq B$  requires that, additionally,  $\tilde{a} \leq \tilde{b}$ .

**Remark 6.** It is interesting to observe that strong and lattice types of orders are related each other by a form of complementarity: for any two intervals  $A, B \in I(\mathbb{R})$

- if  $A, B$  are  $<$ -strong incomparable, then they are  $\subseteq$ -lattice comparable;
- if  $A, B$  are  $\subset$ -strong incomparable, then they are  $\leq$ -lattice comparable.

These facts will be important when considering points of maximum or minimum of an interval-valued function, with respect to the partial orders based on  $\leq_{LU}$  (see Chapter 4 in [4]).

### 3. Interval-valued $gH$ -linear functions of multiple variables

We start this section by first considering a vector-space structure for the set of intervals in terms of midpoint representation (2), useful to define and characterize (appropriate) linearity concepts for  $IV$ -functions on  $\mathbb{R}^n$  (with values in  $I(\mathbb{R})$ ).

The embedding of the space of real intervals into the  $\mathbb{R}^2$  vector space is well known, both in terms of the endpoint and midpoint forms, in particular from the area of Interval Analysis and Set-Valued Analysis.

The main advantage of midpoint representation of intervals is the ‘separation’ between midpoint-value (an element of the field  $\mathbb{R}$  of real numbers) and radius-value, a non negative number giving the half length of the interval. These two components have clear different interpretations which, at least in its very conceptual setting, it is advantageous to keep distinct.

We then start by searching for a vector space structure to represent jointly the two components, midpoint and radius, in terms of the product space of two component vector spaces; as the midpoint of an interval  $X = (\hat{x}; \tilde{x})$  can be any real number  $\hat{x} \in \mathbb{R}$ , it is natural to adopt  $\mathbb{R}$  itself as its underlying vector space (first component space).

For the radius term, due to the sign constraint  $\tilde{x} \geq 0$ , a linear space structure (simply embedded in  $\mathbb{R}$ ) is not allowed. A simple possibility is offered by the observation, suggested as an example in [40] (chapter 1), that the subset  $\mathcal{V}$  of points  $\hat{x} = (x, |x|) \in \mathbb{R}^2$ , with  $x \in \mathbb{R}$ , has naturally the following addition and scalar multiplication for  $\hat{x}, \hat{y} \in \mathcal{V}$  and  $\lambda \in \mathbb{R}$ :

$$\hat{x} + \hat{y} = (x + y, |x + y|) \in \mathcal{V} \text{ and } \lambda \hat{x} = (\lambda x, |\lambda x|) \in \mathcal{V}. \quad (7)$$

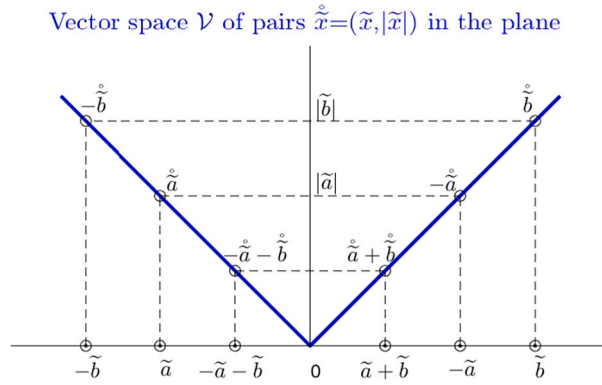
Observe that the addition (7) is not obtained in terms of the addition in  $\mathbb{R}^2$ , indeed, we don’t use the pair  $(x + y, |x| + |y|)$ , which produces a different result when  $x$  and  $y$  have opposite signs.

It is easy to see that  $(\mathcal{V}, +)$  is a commutative group with neutral element  $\hat{0} = (0, 0) \in \mathcal{V}$  and additive inverse  $-\hat{x} = (-x, |x|) \in \mathcal{V}$ , i.e.,  $-(\hat{x}) = \hat{x}$ ; the scalar multiplication satisfies the required properties for a real vector space, i.e.:

$$\lambda(\hat{x} + \hat{y}) = \lambda \hat{x} + \lambda \hat{y}, (\lambda + \mu)\hat{x} = \lambda \hat{x} + \mu \hat{x}, \quad (8)$$

$$(\lambda \mu)\hat{x} = \lambda(\mu \hat{x}), 1\hat{x} = \hat{x}. \quad (9)$$

Clearly,  $\mathcal{V}$  is a unidimensional vector space (represented in Fig. 1), but not a subspace of  $\mathbb{R}^2$  (see [40]). The linear structure of  $\mathcal{V}$  is well described by equality



**Fig. 1.** Vector space  $\mathcal{V}$  of pairs  $\tilde{x} = (\tilde{x}, |\tilde{x}|) \in \mathcal{V}$  with  $\tilde{x} \in \mathbb{R}$ . Elements  $\tilde{a}$  and  $\tilde{b}$ , their opposites  $-\tilde{a}$ ,  $-\tilde{b}$  and additions  $\tilde{a} + \tilde{b}$ ,  $-\tilde{a} - \tilde{b}$  are pictured, as circled points belonging to the blue (V-like) line.

$$\begin{aligned} \lambda \tilde{x} + \mu \tilde{y} &= \lambda(x, |x|) + \mu(y, |y|) = \\ &= (\lambda x, |\lambda x|) + (\mu y, |\mu y|) = \\ &= (\lambda x + \mu y, |\lambda x + \mu y|). \end{aligned}$$

**Remark 7.** Corresponding to positive  $\tilde{x}$  we usually associate  $\tilde{x}$  with a proper 0-symmetric interval  $(0; \tilde{x}) = [-\tilde{x}, \tilde{x}]$  while, for negative  $\tilde{x}$  we have a so called improper (reversed) 0-symmetric interval, well known in classical Interval Analysis literature.

Consider now the product space  $\mathcal{A} = \mathbb{R} \times \mathcal{V}$  where  $\mathbb{R}$  is the vector space of the field  $\mathbb{R}$  on itself (with the ordinary addition and multiplication). The elements of  $\mathcal{A}$  are denoted by  $A = (\hat{a}, \tilde{a})$ , with  $\hat{a}, \tilde{a} \in \mathbb{R}$ ,  $\tilde{a} = (\tilde{a}, |\tilde{a}|) \in \mathcal{V}$  (and similarly for  $B, C \in \mathcal{A}$ ); addition and scalar multiplication in  $\mathcal{A}$  are given, for  $A, B \in \mathcal{A}$  and  $\lambda \in \mathbb{R}$ , by

$$A + B = (\hat{a} + \hat{b}, (\tilde{a} + \tilde{b}, |\tilde{a} + \tilde{b}|)) \in \mathcal{A}, \quad \lambda A = (\lambda \hat{a}, (\lambda \tilde{a}, |\lambda \tilde{a}|)) \in \mathcal{A}.$$

Given  $A, B \in \mathcal{A}$  and  $\alpha, \beta \in \mathbb{R}$ , the linear combination  $C = \alpha A + \beta B \in \mathcal{A}$  is

$$C = (\alpha \hat{a} + \beta \hat{b}, \alpha \tilde{a} + \beta \tilde{b}) = (\alpha \hat{a} + \beta \hat{b}, (\alpha \tilde{a} + \beta \tilde{b}, |\alpha \tilde{a} + \beta \tilde{b}|)).$$

Defining the following equivalence relation on the vector space  $\mathcal{A}$ :

$$A \simeq B \Leftrightarrow \begin{cases} \hat{a} = \hat{b} \\ |\tilde{a}| = |\tilde{b}| \end{cases} \quad (10)$$

the equivalence classes  $[A]_{\simeq} = \{B; B \in \mathcal{A}, B \simeq A\} \subset \mathcal{A}$  are composed, in general, of the two elements  $A$  and  $A'$ , where  $A' = (\hat{a}; -\tilde{a}) = (\hat{a}; (-\tilde{a}, |\tilde{a}|))$  is obtained by the opposite  $-\tilde{a}$  of  $\tilde{a} \in \mathcal{V}$  (with respect to the addition in  $\mathcal{V}$ ) and is such that  $A \simeq A'$  and  $(A')' = A$ , with  $A = A'$  if and only if  $\tilde{a} = 0$ .

The quotient set of the vector space  $\mathcal{A}$ , corresponding to the equivalence relation  $\simeq$  defined in (10), is denoted by  $\mathcal{A}_{\simeq}$ . For  $\alpha \in \mathbb{R}$ , the scalar multiplication on the quotient set  $\mathcal{A}_{\simeq}$  is defined by  $\alpha[A]_{\simeq} = [\alpha A]_{\simeq}$  so that  $\alpha[A]_{\simeq} = \alpha[A']_{\simeq}$ ; indeed, we always have  $(\alpha A)' = \alpha A'$ . Remark that the simple definition of addition in  $\mathcal{A}_{\simeq}$  in terms of the class  $[A + B]_{\simeq}$  is not possible. Indeed, given any  $[A]_{\simeq}, [B]_{\simeq} \in \mathcal{A}_{\simeq}$  with  $[A]_{\simeq} = \{A, A'\}, [B]_{\simeq} = \{B, B'\}$  we have that  $[A + B]_{\simeq} = [A' + B']_{\simeq}$  and  $[A' + B]_{\simeq} = [A + B']_{\simeq}$ , but in general  $[A + B]_{\simeq} \neq [A + B']_{\simeq}$  while  $[B']_{\simeq} = [B]_{\simeq}$  and the addition of two classes cannot be defined as the class  $[A + B]_{\simeq}$  of addition.

On the other hand, the elements of  $\mathcal{A}_{\simeq}$  are subsets of  $\mathcal{A}$  and we can define the addition of classes as a set-addition (in Minkowski sense):

**Definition 8.** Given  $A, B \in \mathcal{A}$  and the corresponding classes  $[A]_{\simeq} = \{A, A'\}, [B]_{\simeq} = \{B, B'\} \in \mathcal{A}_{\simeq}$ , the addition of the two classes is defined to be the following subset of  $\mathcal{A}$

$$[A]_{\simeq} \oplus [B]_{\simeq} = \{A, A'\} \oplus \{B, B'\} = \{A + B, A + B', A' + B, A' + B'\}. \quad (11)$$

The set-addition  $[A]_{\simeq} \oplus [B]_{\simeq}$  is not an  $\simeq$ -equivalence class (unless  $A = A'$  or  $B = B'$ ), but it can be expressed as the set-union of two  $\simeq$ -equivalence classes:

$$[A]_{\simeq} \oplus [B]_{\simeq} = [A + B]_{\simeq} \cup [A + B']_{\simeq} = [A' + B']_{\simeq} \cup [A' + B]_{\simeq}. \quad (12)$$

More precisely, the pair of classes  $[A + B]_{\simeq}$  and  $[A + B']_{\simeq}$  (or, equivalently, the two classes  $[A' + B']_{\simeq}$  and  $[A' + B]_{\simeq}$ ) associated to  $[A]_{\simeq} \oplus [B]_{\simeq}$ , represent two different elements in  $\mathcal{A}_{\simeq}$ , obtained as the union of two different classes  $[C]_{\simeq} \cup [D]_{\simeq}$ , where the components of  $C = (\hat{c}; (\tilde{c}, |\tilde{c}|)) \in \mathcal{A}$  and  $D = (\hat{d}; (\tilde{d}, |\tilde{d}|)) \in \mathcal{A}$  are, respectively,

$$\hat{c} = \hat{a} + \hat{b}, \tilde{c} = \min \{ |\tilde{a} + \tilde{b}|, |\tilde{a} - \tilde{b}| \}, \hat{d} = \hat{a} + \hat{b}, \tilde{d} = \max \{ |\tilde{a} + \tilde{b}|, |\tilde{a} - \tilde{b}| \}. \quad (13)$$

**Remark 9.** It is interesting to observe that the two  $\simeq$ -equivalence classes  $[C]_{\simeq}, [D]_{\simeq}$  are associated, respectively, to the  $gH$ -addition and  $M$ -addition of intervals, i.e.,  $A \oplus_{gH} B \equiv [C]_{\simeq}$  and  $A \oplus_M B \equiv [D]_{\simeq}$ . Correspondingly, if  $A = (\hat{a}; |\tilde{a}|)$ ,  $B = (\hat{b}; |\tilde{b}|)$  with  $\hat{a}, \hat{b}, \tilde{a}, \tilde{b} \in \mathbb{R}$ , then the  $M$ -difference  $A \ominus_M B = A \oplus_M (-B)$  and the  $gH$ -difference  $A \ominus_{gH} B = A \oplus_{gH} (-B)$  are naturally introduced in this setting and given, respectively, by

$$A \ominus_M B = (\hat{a} - \hat{b}; \max \{ |\tilde{a} - \tilde{b}|, |\tilde{a} + \tilde{b}| \}), \quad (14)$$

$$A \ominus_{gH} B = (\hat{a} - \hat{b}; \min \{ |\tilde{a} - \tilde{b}|, |\tilde{a} + \tilde{b}| \}). \quad (15)$$

**Definition 10.** The space  $I(\mathbb{R})$  of real intervals is identified with the quotient set  $\mathcal{A}_{\simeq}$ , i.e., each interval  $A = (\hat{a}, \tilde{a})$ ,  $\tilde{a} \geq 0$  is identified with the equivalence class  $[A]_{\simeq}$ , where we can choose (equivalently)  $A = (\hat{a}; (\tilde{a}, |\tilde{a}|))$  or  $A = (\hat{a}; (-\tilde{a}, |\tilde{a}|))$ .

Each interval  $A = (\hat{a}; \tilde{a}) \in I(\mathbb{R})$  (with  $\tilde{a} \geq 0$ ) has two representations, according to  $[A]_{\simeq} = [A']_{\simeq} = \{A, A'\}$  for appropriate vectors  $A = (\hat{a}; (\tilde{a}, |\tilde{a}|))$  and  $A' = (\hat{a}; (-\tilde{a}, |\tilde{a}|))$ , both belonging to the vector space  $\mathcal{A}$ .

The vector space  $\mathcal{A} = \mathbb{R} \times \mathcal{V}$  with elements  $(\hat{a}, \tilde{a})$  is the product of two unidimensional vector spaces  $\mathbb{R}$  and  $\mathcal{V}$  (both on the field of real numbers  $\mathbb{R}$ ) with  $\hat{a} \in \mathbb{R}$  and  $\tilde{a} \in \mathcal{V}$  (and possibly  $\tilde{a} < 0$ ); as a consequence,  $\mathcal{A}$  has dimension two.

**Remark 11.** In the very recent literature on the calculus for fuzzy-valued functions (see, in particular, [16], [17], [38], [39]) the concept of linearly correlated fuzzy numbers has been deeply analyzed and, for a fixed fuzzy number  $A$ , the fuzzy numbers  $X = qA + r$ , obtained by selecting any  $(q, r) \in \mathbb{R}^2$ , generate a subset  $\mathbb{R}_{\mathcal{F}(A)}$  of fuzzy numbers, called  $A$ -linearly correlated. When the membership function of  $A$  is symmetric around a given value  $\hat{a}$  of its core, the operator  $\Psi_A : \mathbb{R}^2 \rightarrow \mathbb{R}_{\mathcal{F}(A)}$  defined by  $\Psi_A(q, r) = qA + r$  generates the same fuzzy element  $X \in \mathbb{R}_{\mathcal{F}(A)}$  as image of two pairs  $(q, r), (-q, 2q\hat{a} + r)$ ; as detailed, e.g., in [39], an equivalence relation  $\equiv_A$  is established in  $\mathbb{R}^2$ , having the two-elements sets  $[(q, r)]_{\equiv_A} = \{(q, r), (-q, 2q\hat{a} + r)\}$  as equivalence classes.

As suggested by one of the referees, in the case of intervals  $X \in I(\mathbb{R})$ , considered as special fuzzy numbers having all  $\alpha$ -cuts equal to their core, the equivalence relation  $\equiv_A$  is obtained with  $A = [-1, 1]$ .

We clearly have that, denoting the elements of the vector space  $\mathcal{A} = \mathbb{R} \times \mathcal{V}$  as  $X = (\hat{x}; (\tilde{x}, |\tilde{x}|))$  with  $(q, r) \in \mathbb{R}^2$ , the two equivalence classes  $[(r; \tilde{q})]_{\simeq}$  in  $\mathcal{A}_{\simeq}$  and  $[(q, r)]_{\equiv_{[-1, 1]}}$  in  $\mathbb{R}_{\equiv}^2$  represent the same interval  $X \in I(\mathbb{R})$ , respectively, in mid-point notation  $X = (\hat{x}; \tilde{x})$  with  $\hat{x} = r, \tilde{x} = |q| \geq 0$  and, in end-point notation,  $X = [\underline{x}, \bar{x}]$  with  $\underline{x} = r - |q|, \bar{x} = r + |q|$ .

Evidently (and correctly) both settings are able to represent, in their respective cases, all elements of  $I(\mathbb{R})$ .

We could introduce a *canonical base* for  $I(\mathbb{R})$ .

**Definition 12.** Let  $\mathcal{G} = \{G_1, G_2\}$ ,  $G_i = (\hat{g}_i; \tilde{g}_i) \in I(\mathbb{R})$ ,  $i = 1, 2$ , be a set of two intervals. We say that  $\mathcal{G}$  is a base for  $I(\mathbb{R})$  if for any  $A = (\hat{a}; \tilde{a}) \in I(\mathbb{R})$  there exists a pair  $(l_1, l_2) \in \mathbb{R}^2$  such that  $A = (l_1\hat{g}_1 + l_2\hat{g}_2; |l_1\tilde{g}_1 + l_2\tilde{g}_2|)$ ; the pair  $(l_1, l_2)$  is said to represent interval  $A$  with respect to the base  $\mathcal{G}$  and is called a pair of  $\mathcal{G}$ -coordinates of  $A$ . This is denoted as  $A = (l_1, l_2)_{\mathcal{G}}$ .

In the previous definition, the pair  $(l_1, l_2)$  is not necessarily unique, as is seen in the following proposition; this corresponds to the two elements in the equivalence classes  $[A]_{\simeq} = \{A, A'\}$ .

**Proposition 13.** Consider the two intervals  $E_1 = (1; 0)$  and  $E_2 = (0; 1)$ . The ordered set  $\mathcal{E} = \{E_1, E_2\}$  is a base of  $I(\mathbb{R})$  and any interval  $A = (\hat{a}; \tilde{a}) \in I(\mathbb{R})$  with  $\tilde{a} \geq 0$ , is represented by two pairs  $(\hat{a}, \tilde{a})_{\mathcal{E}}$  and  $(\hat{a}, -\tilde{a})_{\mathcal{E}}$  of canonical  $\mathcal{E}$ -coordinates (only one pair if  $\tilde{a} = 0$ ). If  $A$  is identified with the equivalence class of  $A \in \mathcal{A}$ , we can write

$$A = [A]_{\simeq} = \{(\hat{a}, \tilde{a})_{\mathcal{E}}, (\hat{a}, -\tilde{a})_{\mathcal{E}}\}$$

so that both  $(\hat{a}, \tilde{a})_{\mathcal{E}}, (\hat{a}, -\tilde{a})_{\mathcal{E}}$  can be considered as elements of the vector space  $\mathcal{A}$ .

**Proof.** A pair of  $\mathcal{E}$ -coordinates  $(l_1, l_2)_{\mathcal{E}}$  represents interval  $A = (\hat{a}; \tilde{a})$  if and only if  $l_1 = \hat{a}$  and  $|l_2| = \tilde{a}$ . When  $\tilde{a} = 0$ , this is true only for the pair  $(\hat{a}, 0)$ ; when  $\tilde{a} > 0$ , this is true for the two pairs  $(\hat{a}, \tilde{a})$  and  $(\hat{a}, -\tilde{a})$ . It follows that there is a one-to-one correspondence between the set  $\mathcal{A}$  and the set of  $\mathcal{E}$ -coordinates  $(l_1, l_2)_{\mathcal{E}}$ .  $\square$

**Definition 14.** The base  $\mathcal{E} = \{E_1, E_2\} = \{(1; 0), (0; 1)\}$  is said to be the *canonical base* of  $I(\mathbb{R})$ . Proposition 13 implies that, given a representation  $(l_1, l_2)_{\mathcal{E}}$  of interval  $A$  with respect to the canonical base  $\mathcal{E}$ , it follows that  $A = (l_1; |l_2|)$ .

We say that  $A$  is written in *positive signed representation* with respect to the canonical base  $\mathcal{E}$ , if  $l_2 > 0$ ; and in *negative signed representation* if  $l_2 \leq 0$ .

The (canonical) *projection* of  $\mathcal{A}$  onto  $I(\mathbb{R})$  is the mapping  $P : \mathcal{A} \rightarrow I(\mathbb{R})$  defined by  $P((l_1, l_2)_{\mathcal{E}}) = (l_1; |l_2|)$ . The inverse mapping is  $P^{-1}((\hat{a}, \tilde{a})) = \{(\hat{a}, \tilde{a})_{\mathcal{E}}, (\hat{a}, -\tilde{a})_{\mathcal{E}}\}$ .

In general, if  $\{G_1, G_2\}$  is a base,  $G_i = (\hat{g}_i; \tilde{g}_i) \in I(\mathbb{R})$ ,  $i = 1, 2$  and  $(l_1, l_2)^T \in \mathbb{R}^2$ , then the interval  $(l_1 \hat{g}_1 + l_2 \hat{g}_2; |l_1 \tilde{g}_1 + l_2 \tilde{g}_2|)$  is considered as the multiplication  $(l_1, l_2) \odot_{gH} (G_1, G_2)^T$  of the row of real numbers  $(l_1, l_2)$  and column  $(G_1, G_2)^T$  of intervals. The notation will be extended later, to define the  $gH$ -linear combination of intervals and the  $gH$ -based multiplication of a matrix of intervals and a real vector.

**Remark 15.** If  $A = (\hat{a}; \tilde{a})$ ,  $\tilde{a} \geq 0$  is a given interval and  $\mathcal{G} = \{G_1, G_2\}$  is a base,  $G_i = (\hat{g}_i; \tilde{g}_i)$ ,  $i = 1, 2$ , the coordinates of  $A$  with respect to  $\mathcal{G}$  are two real numbers  $l_1, l_2$  such that  $\hat{a} = l_1 \hat{g}_1 + l_2 \hat{g}_2$  and  $\tilde{a} = |l_1 \tilde{g}_1 + l_2 \tilde{g}_2|$ .

Denoting the  $(2, 2)$ -matrix  $M_{\mathcal{G}}$  to have rows  $(\hat{g}_1, \hat{g}_2)$ ,  $(\tilde{g}_1, \tilde{g}_2)$  with  $\det(M_{\mathcal{G}}) > 0$ , we have two systems (1) :  $M_{\mathcal{G}} (l_1, l_2)^T = (\hat{a}, \tilde{a})^T$  or (2) :  $M_{\mathcal{G}} (l_1, l_2)^T = (\hat{a}, -\tilde{a})^T$ . We may choose among the two solutions  $(l_1^{(1)}, l_2^{(1)})$  and  $(l_1^{(2)}, l_2^{(2)})$  according to the sign of  $l_1 \tilde{g}_1 + l_2 \tilde{g}_2$  for the two systems: the two solutions are well defined and, in any case, we have

$$\begin{aligned} l_1^{(1)} &= l_1^{(2)} - 2\tilde{g}_2 \tilde{a} / \det(M_{\mathcal{G}}), \\ l_2^{(1)} &= l_2^{(2)} + 2\tilde{g}_1 \tilde{a} / \det(M_{\mathcal{G}}). \end{aligned}$$

### 3.1. The space of IV-functions and $gH$ -linearity

Considering the elements of  $I(\mathbb{R})$  in terms of equivalence classes of the vector space  $\mathcal{A}$ , has the additional advantage of defining a new concept of linearity in the space on interval-valued functions.

We start with all functions  $F : \mathbb{R}^n \rightarrow \mathcal{A}$ , defined in terms of two arbitrary (ordinary) functions  $\hat{f} : \mathbb{R}^n \rightarrow \mathbb{R}$  and  $\tilde{f} : \mathbb{R}^n \rightarrow \mathbb{R}$  by setting  $F(x) = (\hat{f}(x); (\tilde{f}(x), |\tilde{f}(x)|)) \in \mathcal{A}$ . These functions transform any point  $x \in \mathbb{R}^n$  into the element  $F(x)$  of the vector space  $\mathcal{A}$ .

Such functions are uniquely identified by the pair of functions  $(\hat{f}, \tilde{f})$ .

A function  $F : \mathbb{R}^n \rightarrow \mathcal{A}$  is said to be *linear* if  $\hat{f} : \mathbb{R}^n \rightarrow \mathbb{R}$  and  $\tilde{f} : \mathbb{R}^n \rightarrow \mathbb{R}$  are both linear. The set of all linear functions  $L : \mathbb{R}^n \rightarrow \mathcal{A}$  is denoted by  $\mathcal{L}(\mathbb{R}^n, \mathcal{A})$ . A linear function  $L \in \mathcal{L}(\mathbb{R}^n, \mathcal{A})$  is represented in terms of a pair of vectors  $(\hat{l}, \tilde{l})$  by  $L(x) = (x^T \hat{l}; (x^T \tilde{l}, |x^T \tilde{l}|))$ .

For  $\alpha \in \mathbb{R}$  and  $L, L^{(1)}, L^{(2)} \in \mathcal{L}(\mathbb{R}^n, \mathcal{A})$ , the scalar multiplication and the addition are defined by  $\alpha L(x) = (\alpha x^T \hat{l}; (\alpha x^T \tilde{l}, |\alpha x^T \tilde{l}|))$  and  $L^{(1)}(x) + L^{(2)}(x) = (x^T (\hat{l}^{(1)} + \hat{l}^{(2)}); (x^T (\tilde{l}^{(1)} + \tilde{l}^{(2)}), |x^T (\tilde{l}^{(1)} + \tilde{l}^{(2)})|))$ .

More generally, the equivalence class of a linear combination of  $r$  elements  $A_j = (\hat{a}_j, \tilde{a}_j) \in \mathcal{A}$ ,  $j = 1, 2, \dots, r$ , is the two-elements set

$$\left[ \sum_{j=1}^n x_j A_j \right]_{\simeq} = \left\{ \sum_{j=1}^n x_j A_j, \left( \sum_{j=1}^n x_j A_j \right)' \right\} \quad (16)$$

where

$$\sum_{j=1}^n x_j A_j = \left( \sum_{j=1}^n x_j \hat{a}_j, \left( \sum_{j=1}^n x_j \tilde{a}_j, \left| \sum_{j=1}^n x_j \tilde{a}_j \right| \right) \right)$$

and

$$\left( \sum_{j=1}^n x_j A_j \right)' = \left( \sum_{j=1}^n x_j \hat{a}_j, \left( - \sum_{j=1}^n x_j \tilde{a}_j, \left| - \sum_{j=1}^n x_j \tilde{a}_j \right| \right) \right).$$

According to the identification of  $I(\mathbb{R})$  with the quotient set  $\mathcal{A}_{\simeq}$ , an interval-valued function  $F : \mathbb{R}^n \rightarrow I(\mathbb{R})$  is of the form

$$F(x) = (\hat{f}(x); |\tilde{f}(x)|) = \left[ (\hat{f}(x); (\tilde{f}(x), |\tilde{f}(x)|)) \right]_{\simeq},$$

i.e., using midpoint representation for  $F(x)$ , we can write  $F(x) = (\hat{f}(x); \tilde{f}(x)) \in \mathcal{A}$ , with  $\hat{f}(x), \tilde{f}(x) \in \mathbb{R}$ ,  $\tilde{f}(x) = (\tilde{f}(x), |\tilde{f}(x)|) \in \mathcal{V}$ , so that

$$F(x) = (\hat{f}(x); |\tilde{f}(x)|) := [F(x)]_{\simeq} = \left[ (\hat{f}(x); (\tilde{f}(x), |\tilde{f}(x)|)) \right]_{\simeq}. \quad (17)$$

The space of functions  $F : \mathbb{R}^n \rightarrow \mathcal{A}$  inherits the vector space structure of  $\mathcal{A}$  with addition and scalar multiplication

$$\begin{aligned} \left( \hat{f}(x); \tilde{f}(x) \right) + \left( \hat{g}(x); \tilde{g}(x) \right) &= \left( \hat{f}(x) + \hat{g}(x); \left( \tilde{f}(x) + \tilde{g}(x), \left| \tilde{f}(x) + \tilde{g}(x) \right| \right) \right), \\ \lambda \left( \hat{f}(x); \tilde{f}(x) \right) &= \left( \lambda \hat{f}(x); \left( \lambda \tilde{f}(x), \left| \lambda \tilde{f}(x) \right| \right) \right) \text{ for } \lambda \in \mathbb{R}. \end{aligned}$$

Our definition of linearity for interval-valued functions is the following, based on the vector space structure of  $\mathcal{A}$ , and will be called  $gH$ -linearity.

**Definition 16.** An interval-valued function  $L : \mathbb{R}^n \rightarrow I(\mathbb{R})$  is said to be  $gH$ -linear if there exist two vectors  $\hat{l}, \tilde{l} \in \mathbb{R}^n$  such that for all  $x \in \mathbb{R}^n$  it is  $L(x) = \left( \hat{l}^T x; \left| \tilde{l}^T x \right| \right)$ .

**Proposition 17.** A necessary condition for a function  $L : \mathbb{R}^n \rightarrow I(\mathbb{R})$  to be  $gH$ -linear is that

- (i) it is homogeneous, i.e.,  $L(\lambda x) = \lambda L(x)$  for all  $\lambda \in \mathbb{R}$  and for all  $x \in \mathbb{R}^n$ , and
- (ii) the following inclusion-based order relations are satisfied for all  $x, y \in \mathbb{R}^n$ :

$$L(x) \oplus_{gH} L(y) \subseteq L(x+y) \subseteq L(x) \oplus_M L(y). \quad (18)$$

**Proof.** Both properties are obtained from the definition of  $L(x) = \left( \hat{l}^T x; \left| \tilde{l}^T x \right| \right)$ .

For (i), we have

$$L(\lambda x) = \left( \hat{l}^T (\lambda x); \left| \tilde{l}^T (\lambda x) \right| \right) = \left( \lambda (\hat{l}^T x); |\lambda| \left| \tilde{l}^T x \right| \right) = \lambda \left( \hat{l}^T x; \left| \tilde{l}^T x \right| \right) = \lambda L(x).$$

For the right inequality in (ii), we have

$$L(x+y) = \left( \hat{l}^T x + \hat{l}^T y; \left| \tilde{l}^T x + \tilde{l}^T y \right| \right) \subseteq \left( \hat{l}^T x + \hat{l}^T y; \left| \tilde{l}^T x \right| + \left| \tilde{l}^T y \right| \right) = L(x) \oplus_M L(y).$$

The left inequality in (ii) is deduced analogously, considering the definition  $L(x) \oplus_{gH} L(y) = \left( \hat{l}^T x + \hat{l}^T y; \left| \left| \tilde{l}^T x \right| - \left| \tilde{l}^T y \right| \right| \right)$  and the inequality  $||a| - |b|| \leq |a + b|$ .  $\square$

**Example 18.** Given  $n$  intervals  $A_j = (\hat{a}_j; \tilde{a}_j)$  with  $\hat{a}_j, \tilde{a}_j \in \mathbb{R}$ ,  $\tilde{a}_j \geq 0$  for  $j = 1, 2, \dots, n$ , it is immediate to show that the following IV-function, defined for all  $x \in \mathbb{R}^n$  by

$$F(x) = \left( \sum_{j=1}^n x_j \hat{a}_j; \left| \sum_{j=1}^n x_j \tilde{a}_j \right| \right),$$

is  $gH$ -linear.

The following example shows that conditions (i)-(ii) in the previous Proposition are not sufficient for  $gH$ -linearity.

**Example 19.** Consider the interval-valued function  $F : \mathbb{R}^2 \rightarrow I(\mathbb{R})$ , where  $F(x) = F(x_1, x_2) = x_1[2, 5] \oplus_M x_2[-1, 3]$ , i.e.,  $F$  is the Minkowski combination of two intervals; it is easy to see that  $F$  satisfies the conditions of Proposition 17 but it is not  $gH$ -linear; indeed, we have  $F(x) = \left( \frac{7}{2}x_1 + x_2; \frac{3}{2}|x_1| + 2|x_2| \right)$  and the radius function  $\tilde{f}(x_1, x_2) = \frac{3}{2}|x_1| + 2|x_2|$  cannot be expressed as the absolute value of a linear function of  $(x_1, x_2)$ .

More generally, given  $n$  intervals  $A_j = (\hat{a}_j; \tilde{a}_j)$  with  $\hat{a}_j, \tilde{a}_j \in \mathbb{R}$ ,  $\tilde{a}_j > 0$  for  $j = 1, 2, \dots, n$ , the M-combination  $\sum_{j=1}^n x_j A_j$  of the intervals with coefficients  $x_1, x_2, \dots, x_n$  is not a  $gH$ -linear function, unless  $n = 1$ .

In the rest of the paper, we will make use of the following interval-valued functions, obtained as two different combinations of intervals.

**Definition 20.** Consider an  $n$ -vector  $A = (A_1, A_2, \dots, A_n)^T$  of intervals  $A_j = (\hat{a}_j; \tilde{a}_j)$  and an  $n$ -vector of real coefficients  $x = (x_1, x_2, \dots, x_n)^T$ . We define the  $gH$ -linear combination ( $gH$ -combination for short) as

$$A^T \odot_{gH} x = \bigoplus_{j=1}^n x_j A_j = \left( \sum_{j=1}^n x_j \hat{a}_j; \left| \sum_{j=1}^n x_j \tilde{a}_j \right| \right), \quad (19)$$

while the Minkowski combination (M-combination for short) is

$$A^T \odot_M x = \sum_{j=1}^n x_j A_j = \left( \sum_{j=1}^n x_j \hat{a}_j; \sum_{j=1}^n |x_j| \tilde{a}_j \right). \quad (20)$$

Equivalently, by considering the transposition of vectors, we can write  $A^T \odot_{gH} x = x^T \odot_{gH} A$  and  $A^T \odot_M x = x^T \odot_M A$ .

Remark that, for any set of intervals  $A_j = (\hat{a}_j; \tilde{a}_j)$ ,  $j = 1, 2, \dots, n$  and for any set of real coefficients  $(x_1, x_2, \dots, x_n)$  we have

$$A^T \odot_{gH} x = \bigoplus_{j=1}^n x_j A_j \subseteq \sum_{j=1}^n x_j A_j = A^T \odot_M x \quad (21)$$

and the inclusion reduces to an equality if and only if the nonzero coefficients in the vector  $(x_1, x_2, \dots, x_n)^T$  are all of the same sign.

It is clear that the  $gH$ -combination  $x \rightarrow A^T \odot_{gH} x$  of  $n$  intervals  $A_j = (\hat{a}_j; \tilde{a}_j)$ ,  $\tilde{a}_j \geq 0$ , with any vector  $x \in \mathbb{R}^n$  of coefficients, is a  $gH$ -linear  $IV$ -function on  $\mathbb{R}^n$ , but not all  $gH$ -linear  $IV$ -functions  $x \rightarrow (\hat{l}^T x; |\tilde{l}^T x|)$  can be expressed as  $gH$ -combinations of intervals, i.e., with  $\tilde{l}_j \geq 0$  for all  $j = 1, 2, \dots, n$ .

On the other hand, any  $gH$ -linear  $IV$ -function  $L$ , with interval representation

$$L(x) = \left( \sum_{i=1}^n x_i \hat{l}_i; \left| \sum_{i=1}^n x_i c_i \right| \right),$$

by pair  $(\hat{l}, c)$  of vectors  $\hat{l}, c \in \mathbb{R}^n$ , admits (obviously) also a second representation with pair  $(\hat{l}, -c)$ , given by

$$L(x) = \left( \sum_{i=1}^n x_i \hat{l}_i; \left| \sum_{i=1}^n x_i (-c_i) \right| \right).$$

Making use of the so called *sign pattern matrices* (see [20] and the references therein), it is possible to represent any  $gH$ -linear  $IV$ -function in terms of a modified  $gH$ -combination of an  $n$ -vector  $A$  of intervals by an appropriate invertible (linear) transformation of the  $n$ -vector  $\tilde{a} = (\tilde{a}_1, \tilde{a}_2, \dots, \tilde{a}_n)^T$ . Furthermore, we can choose these transformations so as to avoid the double representation seen above.

**Definition 21.** A *sign pattern matrix* is a matrix whose entries are all from the set  $\{-1, 0, +1\}$ . A square *diagonal sign pattern matrix*  $U = \text{diag}(u_1, \dots, u_n)$  is invertible if all its diagonal elements are in  $\{-1, +1\}$ ; clearly, in that case,  $U^{-1} = U$  and  $U^2 = I$  (the identity matrix).

**Proposition 22.** Given any  $\tilde{w} \in \mathbb{R}^n$ , there exists an invertible diagonal sign pattern matrix  $U = \text{diag}(u_1, u_2, \dots, u_n)$  and a non negative vector  $\tilde{a} = (\tilde{a}_1, \tilde{a}_2, \dots, \tilde{a}_n)^T$ ,  $\tilde{a}_j \geq 0$ , such that  $|\tilde{w}_j| = \tilde{a}_j$ ,  $j = 1, \dots, n$  and  $\tilde{w} = U\tilde{a}$ ,  $\tilde{a} = U\tilde{w}$ ; consequently, for all  $x \in \mathbb{R}^n$ , it is  $|\tilde{w}^T x| = |(U\tilde{a})^T x| = |\tilde{a}^T (Ux)|$ .

**Proof.** To satisfy  $\tilde{a}_j = |\tilde{w}_j|$  it is sufficient to choose

$$u_j = \begin{cases} +1 & \text{if } \tilde{w}_j > 0 \\ -1 & \text{if } \tilde{w}_j \leq 0 \end{cases}$$

so that  $\tilde{a}_j = u_j \tilde{w}_j$  and  $\tilde{w}_j = u_j \tilde{a}_j$ ; then also  $|\sum_{j=1}^n x_j \tilde{w}_j| = |\sum_{j=1}^n x_j u_j \tilde{a}_j|$ .  $\square$

Clearly, if  $U$  is the sign pattern matrix obtained in previous proposition, then also  $V = -U$  has, for all  $x \in \mathbb{R}^n$ ,  $|\tilde{w}^T x| = |(V\tilde{a})^T x|$ .

**Definition 23.** Given any  $n$ -vector of intervals  $A = (A_1, A_2, \dots, A_n)^T$ ,  $A_j = (\hat{a}_j; \tilde{a}_j)$ ,  $\tilde{a}_j \geq 0$ , we say that a nonsingular diagonal sign matrix  $U$  is normalized with respect to  $A$  if and only if  $u_{j^*} = +1$  for the first non zero component  $\tilde{a}_{j^*}$  in vector  $\tilde{a} = (\tilde{a}_1, \tilde{a}_2, \dots, \tilde{a}_n)^T$ . If all components of  $\tilde{a}$  are zero, then  $u_j = -1$  for all  $j$ . The class of such sign matrices, normalized with respect to  $A$ , is denoted by  $N(A)$ .

Observe that the number of elements in  $N(A)$  is  $2^{k_{\tilde{a}}-1}$ , where  $k_{\tilde{a}}$  is the number of non zero elements in  $\tilde{a}$ ; if  $\tilde{a} = 0$ , we assume  $k_{\tilde{a}} = 1$ .

**Proposition 24.** Given any  $n$ -vector of intervals  $A = (A_1, A_2, \dots, A_n)^T$ ,  $A_j = (\hat{a}_j; \tilde{a}_j)$ ,  $\tilde{a}_j \geq 0$ , the set of vectors  $W(A) = \{\tilde{w} \in \mathbb{R}^n : \tilde{w} = U\tilde{a} \text{ for } U \in N(A)\}$  gives a family of  $gH$ -linear functions  $x \rightarrow (\hat{a}^T x; |\tilde{w}^T x|)$ , where  $\tilde{w} \in W(A)$ ; each element of that family is uniquely characterized by an element of the class  $N(A)$ .

**Proof.** Fix any sign matrix  $U = \text{diag}(u_1, u_2, \dots, u_n) \in N(A)$ . Then  $\tilde{w} = U\tilde{a}$  gives the  $gH$ -linear function with  $|\sum_{j=1}^n x_j \tilde{w}_j| = |\sum_{j=1}^n x_j u_j \tilde{a}_j|$ .  $\square$

**Definition 25.** The set of all  $gH$ -linear functions  $L : \mathbb{R}^n \rightarrow I(\mathbb{R})$  is denoted by  $\mathcal{L}_{gH}(\mathbb{R}^n, I(\mathbb{R}))$ . A  $gH$ -linear function  $L \in \mathcal{L}_{gH}(\mathbb{R}^n, I(\mathbb{R}))$  is represented in terms of a pair of vectors  $(\hat{l}, \tilde{l})$  such that, for  $x \in \mathbb{R}^n$ ,  $L(x) = (x^T \hat{l}; |x^T \tilde{l}|)$ . For  $\alpha \in \mathbb{R}$  and  $L, L^{(1)}, L^{(2)} \in \mathcal{L}_{gH}(\mathbb{R}^n, I(\mathbb{R}))$ , the scalar multiplication and the addition are defined by  $\alpha L(x) = (\alpha x^T \hat{l}; |\alpha x^T \tilde{l}|)$  and  $L^{(1)}(x) + L^{(2)}(x) = (x^T(\hat{l}^{(1)} + \hat{l}^{(2)}); |x^T(\tilde{l}^{(1)} + \tilde{l}^{(2)})|)$ .

**Lemma 26.** For any  $n$ -vector of intervals  $(A_1, A_2, \dots, A_n) \in I(\mathbb{R})^n$  there exists a  $gH$ -linear function  $L \in \mathcal{L}_{gH}(\mathbb{R}^n, I(\mathbb{R}))$  such that  $L(e_j) = A_j$ ,  $j = 1, 2, \dots, n$  ( $e_j$  is the  $j$ -th canonical vector of  $\mathbb{R}^n$ ).

**Proof.** Taking  $A_j = (\hat{a}_j; \tilde{a}_j)$ , define  $L(x) = (\sum_{j=1}^n x_j \hat{a}_j; |\sum_{j=1}^n x_j \tilde{a}_j|)$ ;  $L$  is clearly  $gH$ -linear and  $L(e_j) = (\hat{a}_j; |\tilde{a}_j|) = A_j$  for all  $j = 1, 2, \dots, n$ .  $\square$

According to (13),  $gH$ -addition  $F(x) \oplus_{gH} G(x)$  and  $M$ -addition  $F(x) \oplus_M G(x)$  are natural operations for interval-valued functions; e.g., for two  $gH$ -linear functions  $L^{(1)}(x) = (x^T \hat{l}^{(1)}; |x^T \tilde{l}^{(1)}|)$  and  $L^{(2)}(x) = (x^T \hat{l}^{(2)}; |x^T \tilde{l}^{(2)}|)$  we have

$$L^{(1)}(x) \oplus_{gH} L^{(2)}(x) = (x^T(\hat{l}^{(1)} + \hat{l}^{(2)}); \min \{ |x^T(\tilde{l}^{(1)} + \tilde{l}^{(2)})|, |x^T(\tilde{l}^{(1)} - \tilde{l}^{(2)})| \})$$

and

$$L^{(1)}(x) \oplus_M L^{(2)}(x) = (x^T(\hat{l}^{(1)} + \hat{l}^{(2)}); \max \{ |x^T(\tilde{l}^{(1)} + \tilde{l}^{(2)})|, |x^T(\tilde{l}^{(1)} - \tilde{l}^{(2)})| \}).$$

The following property is valid for  $gH$ -linear functions:

**Proposition 27.** For any  $gH$ -linear function  $L : \mathbb{R}^n \rightarrow I(\mathbb{R})$  we have, for all  $\alpha, \beta \in \mathbb{R}$  and all  $x, y \in \mathbb{R}^n$ :

$$\alpha L(x) \oplus_{gH} \beta L(y) \subseteq L(\alpha x + \beta y) \subseteq \alpha L(x) \oplus_M \beta L(y).$$

**Proof.** Denote  $L(x) = (l^T x; |w^T x|)$ ; then

$$\alpha L(x) \oplus_{gH} \beta L(y) = (\alpha l^T x + \beta l^T y; | \alpha w^T x | - | \beta w^T y | ),$$

$$L(\alpha x + \beta y) = (\alpha l^T x + \beta l^T y; | \alpha w^T x + \beta w^T y | ) \text{ and}$$

$$\alpha L(x) \oplus_M \beta L(y) = (\alpha l^T x + \beta l^T y; | \alpha w^T x | + | \beta w^T y | );$$

from the inequalities  $| |a| - |b| | \leq |a - b| \leq |a| + |b|$  applied to  $a = \alpha w^T x$ ,  $b = \beta w^T y$ , we immediately have the two required inclusions.  $\square$

**Remark 28.** With the help of operations  $\oplus_{gH}, \oplus_M$  defined in Remark 9, we say that an  $IV$ -function  $F : \mathbb{R}^n \rightarrow I(\mathbb{R})$  is

(a)  $\oplus_M$ -subadditive (i.e., with respect to Minkowski  $M$ -addition) if

$$F(x + y) \subseteq F(x) \oplus_M F(y) \text{ for all } x, y \in \mathbb{R}^n, \quad (22)$$

(b)  $\oplus_{gH}$ -superadditive (i.e., with respect to  $gH$ -addition) if

$$F(x) \oplus_{gH} F(y) \subseteq F(x + y) \text{ for all } x, y \in \mathbb{R}^n. \quad (23)$$

Using this terminology, a necessary condition for  $F$  to be  $gH$ -linear is that it is homogeneous (i.e.,  $F(\lambda x) = \lambda F(x)$  for all  $\lambda \in \mathbb{R}$  and  $x \in \mathbb{R}^n$ ),  $\oplus_M$ -subadditive and  $\oplus_{gH}$ -superadditive.

#### 4. Interval Fréchet and Gateaux $gH$ -differentiability

We have now all elements to define Fréchet and Gateaux type differentiability for an interval-valued function and, in particular, the Fréchet (interval-valued)  $gH$ -differential function.

In the rest of the paper we will denote an  $IV$ -function  $F : K \subseteq \mathbb{R}^n \rightarrow I(\mathbb{R})$  as  $F(x) = (\hat{f}(x); \tilde{f}(x))$ , with  $\hat{f} : K \rightarrow \mathbb{R}$ ,  $\tilde{f} : K \rightarrow \mathbb{R}$ ,  $\tilde{f}(x) \geq 0$  for all  $x \in K$ .

Frequently, the radius component  $\hat{f}(x)$  of  $F(x)$  will be considered to be the absolute value  $|\tilde{f}_w(x)|$  of a function  $\tilde{f}_w : K \rightarrow \mathbb{R}$  with possibly  $\tilde{f}_w(x) \leq 0$  in a subset of points  $x \in K$ .

Similarly, for  $gH$ -linear functions we will frequently write  $L = (\hat{l}; |\tilde{w}|)$  where  $w : \mathbb{R}^n \rightarrow \mathbb{R}$  is a standard linear function, i.e.,  $\tilde{l}(h) = |\tilde{w}(h)|$  for all  $h \in \mathbb{R}^n$ .

We will denote the terms of linear function  $\tilde{w}(h)$ ,  $h \in \mathbb{R}^n$  as a scalar product  $\tilde{w}(h) = \tilde{w}^T h$ , where  $\tilde{w} = (\tilde{w}_1, \tilde{w}_2, \dots, \tilde{w}_n) \in \mathbb{R}^n$  is a vector of coefficients.

In general, to account for a  $gH$ -linear function  $L = (\hat{l}; \tilde{l}) : \mathbb{R}^n \rightarrow I(\mathbb{R})$ ,  $\tilde{l} \geq 0$ , be depending on a point  $x^{(0)}$ , we will denote it as  $L_{x^{(0)}}(h) = (\hat{l}_{x^{(0)}}(h); \tilde{l}_{x^{(0)}}(h)) = (\hat{l}^T h; |\tilde{w}^T h|)$ ; when there is no risk of confusion, we omit explicit reference to  $x^{(0)}$  and write simply  $L(h) = (\hat{l}(h); \tilde{l}(h)) = (\hat{l}^T h; |\tilde{w}^T h|)$ .

**Definition 29.** Let  $F : K \subseteq \mathbb{R}^n \rightarrow I(\mathbb{R})$ ,  $F(x) = (\hat{f}(x); \tilde{f}(x)) = [\hat{f}(x) - \tilde{f}(x), \hat{f}(x) + \tilde{f}(x)]$  and let  $x^{(0)} \in K$  such that  $x^{(0)} + h \in K$ , for all  $h \in \mathbb{R}^n$  with  $\|h\| < \delta$  for a given  $\delta > 0$ . We say that  $F$  is *Fréchet  $gH$ -differentiable* at  $x^{(0)}$  if and only if there exists a (continuous)  $gH$ -linear function  $L_{x^{(0)}} : \mathbb{R}^n \rightarrow I(\mathbb{R})$  such that

$$\lim_{\|h\| \rightarrow 0} \frac{\| (F(x^{(0)} + h) \ominus_{gH} F(x^{(0)})) \ominus_{gH} L_{x^{(0)}}(h) \|_H}{\|h\|} = 0 \quad (24)$$

The interval-valued function  $L_{x^{(0)}} : h \mapsto L_{x^{(0)}}(h)$ ,  $h \in \mathbb{R}^n$ , is called the *Fréchet  $gH$ -derivative* (or *Fréchet  $gH$ -differential function*) of  $F$  at  $x^{(0)}$ , denoted as  $D_{gH}F(x^{(0)})$ , i.e.,  $D_{gH}F(x^{(0)})(h) = L_{x^{(0)}}(h)$  for all  $h \in \mathbb{R}^n$ .

$F$  is said *Fréchet  $gH$ -differentiable* on a set  $K \subseteq \mathbb{R}^n$  if it is *Fréchet  $gH$ -differentiable* at any  $x^{(0)} \in K$ .

Some motivations and explanation for our choice of  $gH$ -linearity as the “target class” of functions  $L_{x^{(0)}} : \mathbb{R}^n \rightarrow I(\mathbb{R})$  used in the above definition, are important before examining its properties.

To this matter, we advance that if we replace the requirement of the  $gH$ -linearity in Definition 29 by ‘linearity’ (in Minkowski addition and scalar multiplication sense), we can find some non expected issues (for instance, non existence), as we comment in a next Section 4.3.

To have *Fréchet  $gH$ -differentiability*, it is clear that the following two (standard) limits must be valid as necessary and sufficient conditions for (24):

$$\lim_{h \rightarrow 0} \frac{(\hat{f}(x^{(0)} + h) - \hat{f}(x^{(0)})) - \hat{l}_{x^{(0)}}(h)}{\|h\|} = 0, \text{ and} \quad (25)$$

$$\lim_{h \rightarrow 0} \frac{|\tilde{f}(x^{(0)} + h) - \tilde{f}(x^{(0)})| - \tilde{l}_{x^{(0)}}(h)}{\|h\|} = 0. \quad (26)$$

With respect to the midpoint component  $\hat{f}$ , a standard real-valued function, it is natural to require that  $h \mapsto \hat{l}_{x^{(0)}}(h)$ ,  $h \in \mathbb{R}^n$ , be a standard linear function, in the form  $\hat{l}(h) = \hat{l}_1 h_1 + \hat{l}_2 h_2 + \dots + \hat{l}_n h_n$  with  $\hat{l}_j \in \mathbb{R}$  for  $j = 1, 2, \dots, n$ .

Let’s examine now the radius component  $\tilde{l}(h)$ , assumed here to be the absolute value of a linear function,  $\tilde{l}(h) = |\sum_{j=1}^n \tilde{w}_j h_j|$ , which is a “special” case of the class of Minkowski functionals, also called *gauge functions* in Functional Analysis (see, e.g., [30], [37]) and in Convex-Variational Analysis (see [33], [34]). See also [7] for a recent unifying view.

Consider first the case where the midpoint component  $\tilde{f}$  is itself *Fréchet differentiable* in the ordinary sense with linear differential function  $h \mapsto \tilde{w}(h)$ , not necessarily  $\tilde{w}(h) \geq 0$ ; then, from

$$\lim_{h \rightarrow 0} \frac{(\tilde{f}(x^{(0)} + h) - \tilde{f}(x^{(0)})) - \tilde{w}(h)}{\|h\|} = 0, \quad (27)$$

by applying the second triangular inequality  $||\alpha| - |\beta|| \leq |\alpha - \beta|$ , it follows that (26) is satisfied with  $\tilde{l}_{x^{(0)}}(h) = |\tilde{w}(h)|$ .

When  $\tilde{f}(x)$  is not differentiable, it is then natural we address attention to functions  $\tilde{l}(h) \geq 0$  that can be obtained as some form of combination of  $n$  intervals  $A_j = (\hat{a}_j; \tilde{a}_j) = (\hat{a}_j; \tilde{w}_j)$ , where each  $\tilde{w}_j \in \mathbb{R}$  (in particular its sign, as we will see) are relevant to back-reconstruction of function  $\tilde{l}(h) \geq 0$  from partial derivatives.

This connects to the important property that (see, e.g., chapter 7 in [30]) the linear functions  $w(h)$  on a real vector space (in particular  $\mathbb{R}^n$ ) are the *minimal* elements of the class of sub-linear functions (i.e., no sub-linear function  $p(h)$  exists with  $p(h) < w(h)$  for at least one point  $h$ ); this is important to ensure unicity of the *Fréchet  $gH$ -differential function* and to distinguish it from other possible tangency conditions (see below). Furthermore, for any linear function  $w(h)$  the map  $h \mapsto |w(h)|$  is always non-negative, absolutely homogeneous and sub-additive (i.e., a *seminorm* function); on the other hand, by the application of the Hahn-Banach Theorem (see Theorem 7.3.4 in [30]), we know that for any seminorm  $p : \mathbb{R}^n \rightarrow \mathbb{R}^+ \cup \{0\}$  there exists a linear function  $w : \mathbb{R}^n \rightarrow \mathbb{R}$  such that  $|w(h)| \leq p(h)$  for all  $h \in \mathbb{R}^n$  and  $w$  is continuous if and only if so is also  $p$  (Theorem 5.7.3 in [30]).

Precisely these facts suggest to consider the class of  $gH$ -linear *IV*-functions  $L(h) = (\hat{l}^T h; |\tilde{w}^T h|)$ ,  $h \in \mathbb{R}^n$  with  $\tilde{w} \in \mathbb{R}^n$  as appropriate candidates for our definition of *Fréchet  $gH$ -differentiability* for *IV*-functions.

We can characterize our definition of *Fréchet  $gH$ -differentiability* by first defining a concept of “absolute” differentiability for an ordinary function.

**Definition 30.** Let  $f : K \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ ,  $f : x \mapsto f(x)$ , be any given real-valued function and let  $x^{(0)} \in K$  such that  $x^{(0)} + h \in K$ , for all  $h \in \mathbb{R}^n$  with  $\|h\| < \delta$  for a given  $\delta > 0$ . We say that  $f$  is *Fréchet absolute differentiable* (abs-differentiable, for short) at  $x^{(0)}$  if and only if there exists a (continuous) linear function  $w_{x^{(0)}} : \mathbb{R}^n \rightarrow \mathbb{R}$  such that

$$\lim_{\|h\| \rightarrow 0} \frac{\left| \left( |f(x^{(0)} + h)| - |f(x^{(0)})| \right) - |w_{x^{(0)}}(h)| \right|}{\|h\|} = 0 \quad (28)$$

The linear real-valued function  $w_{x^{(0)}} : h \rightarrow w_{x^{(0)}}(h)$ ,  $h \in \mathbb{R}^n$ , is called the *Fréchet abs-differential* of  $f$  at  $x^{(0)}$ , denoted as  $D_{abs}f(x^{(0)}) : \mathbb{R}^n \rightarrow \mathbb{R}$ , i.e.,  $D_{abs}f(x^{(0)})(h) = w_{x^{(0)}}(h)$  for all  $h \in \mathbb{R}^n$ ; there exists a function  $r(h)$  with  $\lim_{h \rightarrow 0} r(h) = 0$  such that  $\left| \left( |f(x^{(0)} + h)| - |f(x^{(0)})| \right) - |w_{x^{(0)}}(h)| \right| = |r(h)| \|h\|$ .

Additionally, if for a given  $j \in \{1, 2, \dots, n\}$  there exists  $w_{x^{(0)},j} \in \mathbb{R}$  (depending on  $x^{(0)}$  and  $j$ ) and a function  $r_{x^{(0)},j} : \mathbb{R} \rightarrow \mathbb{R}$  with  $\lim_{t \rightarrow 0} r_{x^{(0)},j}(t) = 0$  satisfying

$$\left| \left( |f(x^{(0)} + te_j)| - |f(x^{(0)})| \right) - |w_{x^{(0)},j}t| \right| = |t| r_{x^{(0)},j}(t),$$

we say that  $w_{x^{(0)},j}$  is a *partial abs-derivative* of  $f$  at  $x^{(0)}$  with respect to  $x_j$  (or in the canonical direction  $e_j$ ).

If, for some index  $j \in \{1, 2, \dots, n\}$ , a number  $w_{x^{(0)},j} \neq 0 \in \mathbb{R}$  is a partial abs-derivative of  $f$ , then also  $-w_{x^{(0)},j} \in \mathbb{R}$  satisfies the same property; but it is easy to prove that the partial abs-derivatives are essentially unique in the sense that no other different numbers can satisfy the same property.

**Proposition 31.** *If  $f : K \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$  is abs-differentiable at  $x^{(0)} \in K$  with abs-differential function  $w(h)$ , then  $w$  is essentially unique, in the sense that if  $w_1, w_2 : \mathbb{R}^n \rightarrow \mathbb{R}$  satisfy equation (28), then  $|w_1(h)| = |w_2(h)|$  for all  $h \in \mathbb{R}^n$ .*

**Proof.** Suppose there is  $h_0 \in \mathbb{R}^n$  with  $|w_1(h_0)| \neq |w_2(h_0)|$ ; let  $0 \neq t_0 \in \mathbb{R}$  be such that  $x^{(0)} + th_0 \in K$  for all  $0 < t \leq t_0$ , then

$$\begin{aligned} \left| |w_1(h_0)| - |w_2(h_0)| \right| &= \left| \frac{1}{|t|} (|w_1(th_0)| - |w_2(th_0)|) \right| = \\ &= \left| \frac{1}{\|th_0\|} (|w_1(th_0)| - |w_2(th_0)|) \right| \|h_0\|. \end{aligned}$$

Then, by adding/subtracting  $\left| (|f(x^{(0)} + th_0)| - |f(x^{(0)})|) - |w_1(th_0)| \right|$  and applying triangle inequality, we obtain

$$\begin{aligned} \left| |w_1(h_0)| - |w_2(h_0)| \right| &\leq \frac{1}{\|th_0\|} \left| \left( |f(x^{(0)} + th_0)| - |f(x^{(0)})| \right) - |w_1(th_0)| \right| \|h_0\| \\ &\quad + \frac{1}{\|th_0\|} \left| \left( |f(x^{(0)} + th_0)| - |f(x^{(0)})| \right) - |w_2(th_0)| \right| \|h_0\|; \end{aligned}$$

the two terms on the right side, taking the limit for  $t \rightarrow 0$ , tend to zero by abs-differentiability and consequently we get the contradiction  $\left| |w_1(h_0)| - |w_2(h_0)| \right| = 0$ .  $\square$

An immediate connection with standard Fréchet differentiability is the following property:

**Proposition 32.** *If  $f : K \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$  is such that its absolute value  $|f| : K \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$  is Fréchet differentiable at  $x^{(0)} \in K$  with differential function  $D|f|(x^{(0)})$ , then  $f$  is abs-differentiable at  $x^{(0)}$  with the same differential (i.e.,  $D_{abs}f(x^{(0)}) = D|f|(x^{(0)})$ ).*

*In particular, if  $f : K \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^+ \cup 0$  is differentiable then it is also abs-differentiable.*

**Proof.** From the differentiability of  $|f|$  at  $x^{(0)}$  there exists a linear function  $w(h)$  such that

$$\lim_{h \rightarrow 0} \frac{\left| |f(x^{(0)} + h)| - |f(x^{(0)})| - w(h) \right|}{\|h\|} = 0;$$

by applying the second triangular inequality, we have

$$0 \leq \left| \frac{\left| \left( |f(x^{(0)} + h)| - |f(x^{(0)})| \right) - |w(h)| \right|}{\|h\|} \right| \leq \left| \frac{|f(x^{(0)} + h)| - |f(x^{(0)})| - w(h)}{\|h\|} \right|$$

and the limit required in (28) is satisfied with  $\tilde{l}_{x^{(0)}}(h) = |w(h)|$ .  $\square$

Observe that abs-differentiability of an ordinary function does not imply its differentiability, as is the case when a partial derivative does not exist because the left and right partial derivatives exist but have opposite values. It is important to consider that, on the other hand, abs-differentiability does not imply ordinary Fréchet differentiability and is adopted here to mark some distinction between the requirements on midpoint and radius components.

**Remark 33.** It does not seem possible to establish general sufficient conditions for abs-differentiability of  $f$ , similar to the well known one based on the class of  $C^{(1)}$  functions.

A useful fact is that for a function  $f(\cdot)$  to be abs-differentiable it is necessary that left and right partial derivatives need to exist and have the same absolute value; in particular, if there exists  $j$  such that  $\left| \frac{\partial f(x^{(0)})}{\partial x_j^-} \right| \neq \left| \frac{\partial f(x^{(0)})}{\partial x_j^+} \right|$ , then  $f$  cannot be abs-differentiable at  $x^{(0)}$ .

With the help of abs-differentiability, we can express an equivalent definition of Fréchet interval-valued  $gH$ -differentiability (the proof follows immediately from the two definitions):

**Theorem 34.** The IV-function  $F : K \subseteq \mathbb{R}^n \rightarrow I(\mathbb{R})$ ,  $F(x) = (\hat{f}(x); \tilde{f}(x))$  is Fréchet  $gH$ -differentiable at a point  $x^{(0)} \in K$ , if and only if,  $\hat{f}$  and  $\tilde{f}$  are (respectively) Fréchet differentiable and abs-differentiable at  $x^{(0)}$ .

Similarly to Fréchet  $gH$ -differentiability (usually called *strong*), we can define *weak* (or Gateaux-type)  $gH$ -differentiability:

**Definition 35.** Let  $F : K \subseteq \mathbb{R}^n \rightarrow I(\mathbb{R})$ ,  $F(x) = (\hat{f}(x); \tilde{f}(x)) = [\hat{f}(x) - \tilde{f}(x), \hat{f}(x) + \tilde{f}(x)]$  and let  $x^{(0)} \in K$  such that  $x^{(0)} + h \in K$ . We say that  $F$  is Gateaux  $gH$ -differentiable at  $x^{(0)}$  if and only if there exists a (continuous)  $gH$ -linear function  $L : \mathbb{R}^n \rightarrow I(\mathbb{R})$ , depending only on  $x^{(0)}$  such that, for all  $h \in \mathbb{R}^n$  the following limit exists:

$$\lim_{t \rightarrow 0} \frac{(F(x^{(0)} + th) \ominus_{gH} F(x^{(0)}))}{t} = L(h) \quad (29)$$

The  $gH$ -linear interval-valued function  $L : h \rightarrow L(h)$  on  $\mathbb{R}^n$  is called the Gateaux  $gH$ -derivative of  $F$  at  $x^{(0)}$ . Clearly, if it exists, it is unique (by unicity of the limit for all  $h$ ).

Any Fréchet  $gH$ -differentiable IV-function is Gateaux  $gH$ -differentiable; the vice versa is not true as, in particular, Gateaux  $gH$ -differentiability does not imply continuity.

#### 4.1. Basic properties of Fréchet and Gateaux $gH$ -differentiability

The following are fundamental properties of Fréchet  $gH$ -differentiability.

**Theorem 36.** Let  $F : K \subseteq \mathbb{R}^n \rightarrow I(\mathbb{R})$ ,  $F(x) = (\hat{f}(x); \tilde{f}(x))$  and let  $x^{(0)} \in K$ . If  $F$  is Fréchet  $gH$ -differentiable at  $x^{(0)}$ , then  $F$  is continuous at  $x^{(0)}$ .

**Proof.** We clearly have

$$\lim_{\|h\| \rightarrow 0} \| (F(x^{(0)} + h) \ominus_{gH} F(x^{(0)})) \ominus_{gH} L_{x^{(0)}}(h) \|_H = 0, \quad (30)$$

i.e.,

$$\lim_{\|h\| \rightarrow 0} F(x^{(0)} + h) \ominus_{gH} F(x^{(0)}) = L_{x^{(0)}}(0) = (0; 0), \quad (31)$$

i.e.,  $\lim_{\|h\| \rightarrow 0} F(x^{(0)} + h) = F(x^{(0)})$ .  $\square$

The following Theorem ensures that  $gH$ -linear IV-functions are Fréchet  $gH$ -differentiable and preserve one of the basic properties of classical differentiability: the total differential of a linear function  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  exists at any point and coincides with the  $f$  itself.

**Theorem 37.** Let  $F : \mathbb{R}^n \rightarrow I(\mathbb{R})$ ,  $F(x) = (\hat{f}(x); \tilde{f}(x))$  be  $gH$ -linear. Then, at any point  $x \in \mathbb{R}^n$ ,  $F$  is Fréchet  $gH$ -differentiable and its  $gH$ -differential is exactly the function  $D_{gH}F(x)(h) = F(h)$  for all  $h \in \mathbb{R}^n$  and all  $x \in \mathbb{R}^n$ .

**Proof.** Let  $\hat{f}(x) = \sum_{i=1}^n x_i \hat{c}_i$  and  $\tilde{f}(x) = |\sum_{i=1}^n x_i c_i|$  with given (fixed)  $\hat{c}_i, c_i \in \mathbb{R}$ . We then have the following equalities for all  $x, h \in \mathbb{R}^n$  with  $h \neq 0$ :

$$\begin{aligned} F(x) &= \left( \sum_{i=1}^n x_i \hat{c}_i; \left| \sum_{i=1}^n x_i c_i \right| \right) \\ F(x+h) &= \left( \sum_{i=1}^n (x_i + h_i) \hat{c}_i; \left| \sum_{i=1}^n (x_i + h_i) c_i \right| \right) \end{aligned}$$

$$F(h) = \left( \sum_{i=1}^n h_i \hat{c}_i; \left| \sum_{i=1}^n h_i c_i \right| \right)$$

so that, the second component being the absolute value of the difference of two absolute values,

$$F(x+h) \ominus_{gH} F(x) = \left( \sum_{i=1}^n h_i \hat{c}_i; \left| \sum_{i=1}^n (x_i + h_i) c_i \right| - \left| \sum_{i=1}^n x_i c_i \right| \right)$$

and, denoting the interval  $A_{x,h} = (F(x+h) \ominus_{gH} F(x)) \ominus_{gH} F(h)$  with  $A_{x,h} = (\hat{A}_{x,h}; \tilde{A}_{x,h})$  in midpoint notation, it follows that

$$\begin{cases} \hat{A}_{x,h} = 0 \\ \tilde{A}_{x,h} = \left| \left| \sum_{i=1}^n (x_i + h_i) c_i \right| - \left| \sum_{i=1}^n x_i c_i \right| \right| - \left| \sum_{i=1}^n h_i c_i \right| \end{cases};$$

on the other hand, from the fact that  $\hat{A}_{x,h} = 0$ , it is  $\|A_{x,h}\|_H = \tilde{A}_{x,h}$  with

$$\tilde{A}_{x,h} = \left| \left| \sum_{i=1}^n x_i c_i + \sum_{i=1}^n h_i c_i \right| - \left| \sum_{i=1}^n x_i c_i \right| \right| - \left| \sum_{i=1}^n h_i c_i \right|$$

and consequently

$$\frac{\|(F(x+h) \ominus_{gH} F(x)) \ominus_{gH} F(h)\|_H}{\|h\|} = \frac{\tilde{A}_{x,h}}{\|h\|}.$$

Considering that the vector  $x \in \mathbb{R}^n$  is fixed, we have three cases for the sign of the quantity  $\sum_{i=1}^n x_i c_i$  (not depending on  $h$ ):

1) if  $\sum_{i=1}^n x_i c_i = 0$ , then

$$\tilde{A}_{x,h} = \left| \left| 0 + \sum_{i=1}^n h_i c_i \right| - |0| - \left| \sum_{i=1}^n h_i c_i \right| \right| = 0$$

so that  $\frac{\tilde{A}_{x,h}}{\|h\|} = 0$  for all  $h \neq 0$ ;

2) if  $\sum_{i=1}^n x_i c_i \geq \rho > 0$ , then for sufficiently small  $\|h\|$ , it will be  $\sum_{i=1}^n x_i c_i + \sum_{i=1}^n h_i c_i \geq 0$  so that

$$\begin{aligned} \tilde{A}_{x,h} &= \left| \left| \sum_{i=1}^n x_i c_i + \sum_{i=1}^n h_i c_i - \sum_{i=1}^n x_i c_i \right| - \left| \sum_{i=1}^n h_i c_i \right| \right| \\ &= \left| \left| \sum_{i=1}^n h_i c_i \right| - \left| \sum_{i=1}^n h_i c_i \right| \right| = 0; \end{aligned}$$

3) if  $\sum_{i=1}^n x_i c_i \leq \rho < 0$ , then for sufficiently small  $\|h\|$ , it will be  $\sum_{i=1}^n x_i c_i + \sum_{i=1}^n h_i c_i \leq 0$  so that

$$\begin{aligned} \tilde{A}_{x,h} &= \left| \left| -\sum_{i=1}^n x_i c_i - \sum_{i=1}^n h_i c_i + \sum_{i=1}^n x_i c_i \right| - \left| \sum_{i=1}^n h_i c_i \right| \right| \\ &= \left| \left| -\sum_{i=1}^n h_i c_i \right| - \left| \sum_{i=1}^n h_i c_i \right| \right| = 0. \end{aligned}$$

Then, in all cases,  $\tilde{A}_{x,h} = 0$  for all  $h \in \mathbb{R}^n$  with sufficiently small  $\|h\|$  and for all  $x \in \mathbb{R}^n$ ; we conclude that  $\lim_{\|h\| \rightarrow 0} \frac{\|(F(x+h) \ominus_{gH} F(x)) \ominus_{gH} F(h)\|_H}{\|h\|} = 0$  and  $F$  is Fréchet  $gH$ -differentiable with differential  $D_{gH}F(x)(h) = F(h)$ .  $\square$

In a recent paper (see [45]), the following definition of  $gH$ -differentiability has been given; it implicitly suggested our idea to define linearity in the context of interval-valued functions.

**Definition 38.** (see [45]) Let  $F : K \subseteq \mathbb{R}^n \rightarrow I(\mathbb{R})$ ,  $F(x) = (\hat{f}(x); \tilde{f}(x)) = [\hat{f}(x) - \tilde{f}(x), \hat{f}(x) + \tilde{f}(x)]$  and let  $x^{(0)} \in K$  such that  $x^{(0)} + h \in K$ , for all  $h \in \mathbb{R}^n$  with  $\|h\| < \delta$  for a given  $\delta > 0$ . We say that  $F$  is  $gH$ -differentiable at  $x^{(0)}$  if and only if there exist two vectors  $\hat{w}, \tilde{w} \in \mathbb{R}^n$ ,  $\hat{w} = (\hat{w}_1, \dots, \hat{w}_n)$ ,  $\tilde{w} = (\tilde{w}_1, \dots, \tilde{w}_n)$  and two functions  $\hat{\varepsilon}(h)$ ,  $\tilde{\varepsilon}(h)$  with  $\lim_{h \rightarrow 0} \hat{\varepsilon}(h) = \lim_{h \rightarrow 0} \tilde{\varepsilon}(h) = 0$ , such that, for all  $h \neq 0$ ,

$$\hat{f}(x^{(0)} + h) - \hat{f}(x^{(0)}) = \sum_{j=1}^n h_j \hat{w}_j + \|h\| \hat{\varepsilon}(h), \quad (32)$$

$$\left| \tilde{f}(x^{(0)} + h) - \tilde{f}(x^{(0)}) \right| = \left| \sum_{j=1}^n h_j \tilde{w}_j + \|h\| \tilde{\varepsilon}(h) \right|. \quad (33)$$

The interval-valued function  $F'_{gH}(x^{(0)}; \cdot) : \mathbb{R}^n \rightarrow I(\mathbb{R})$  defined, for  $h = (h_1, \dots, h_n) \in \mathbb{R}^n$ , by

$$F'_{gH}(x^{(0)}; h) = \left( \sum_{j=1}^n h_j \hat{w}_j; \left| \sum_{j=1}^n h_j \tilde{w}_j \right| \right) \quad (34)$$

is called the total  $gH$ -derivative of  $F$  at  $x^{(0)}$ .

It is clear that the total  $gH$ -derivative and the Fréchet  $gH$ -differential function in Definition 29 coincide. What is important to remark here is that the setting of  $gH$ -linear  $IV$ -functions was not explicitly considered in [45].

In fact, in Definitions 29 and 38, both functions  $L_{x^{(0)}}(h)$  in (24) and  $F'_{gH}(x^{(0)}; h)$  in (34) belong to the family of  $gH$ -linear functions and they coincide, as we see in the proof of the next Proposition.

**Proposition 39.** Let  $F : K \subseteq \mathbb{R}^n \rightarrow I(\mathbb{R})$ ,  $F(x) = (\hat{f}(x); \tilde{f}(x))$ . Then,  $F(x)$  is Fréchet  $gH$ -differentiable at  $x^{(0)} \in K$ , according to Definition 29, if and only if the total  $gH$ -derivative at  $x^{(0)}$  exists (according to Definition 38) and

$$F'_{gH}(x^{(0)}; h) = D_{gH}F(x^{(0)})(h) \text{ for all } h \in \mathbb{R}^n. \quad (35)$$

**Proof.** In both definitions, the midpoint function  $\hat{f}$  is Fréchet differentiable in the ordinary sense and nothing more is to be proved for it. It remains to prove that equation (33) is equivalent to abs-differentiability of function  $\tilde{f}(x) \geq 0$  at point  $x^{(0)}$  with  $w_{x^{(0)}}(h) = \sum_{j=1}^n h_j \tilde{w}_j$ . Take  $h \neq 0$  so that  $\|h\| > 0$ ; define the function  $\tilde{r}(h) = \frac{|w_{x^{(0)}}(h) + \|h\| \tilde{e}(h)| - |w_{x^{(0)}}(h)|}{\|h\|}$ , such that  $|w_{x^{(0)}}(h) + \|h\| \tilde{e}(h)| = |w_{x^{(0)}}(h)| + \|h\| \tilde{r}(h)$ . By (33) we necessarily have that  $\|h\| \tilde{r}(h) \rightarrow 0$  as  $h \rightarrow 0$ . On the other hand, by applying the second triangular inequality, we also have  $0 \leq |\tilde{r}(h)| \leq |\tilde{e}(h)| \rightarrow 0$  as  $h \rightarrow 0$ . We conclude that

$$\frac{|\tilde{f}(x^{(0)} + h) - \tilde{f}(x^{(0)})| - \left| \sum_{j=1}^n h_j \tilde{w}_j \right|}{\|h\|} = \tilde{r}(h) \rightarrow 0 \text{ as } h \rightarrow 0$$

and  $\tilde{f}$  is abs-differentiable at  $x^{(0)}$ .  $\square$

**Remark 40.** The authors of [18] define a “linearity” concept for a function  $L : \mathcal{X} \rightarrow I(\mathbb{R})$ , with  $L(x) = [\underline{L}(x), \overline{L}(x)]$  for  $x \in \mathcal{X}$ ,  $\mathcal{X}$  being a linear subspace of  $\mathbb{R}^n$ , by requiring that

- (i)  $L(\lambda x) = \lambda L(x)$  for all  $x \in \mathcal{X}$  and  $\lambda \in \mathbb{R}$ ;
- (ii) for all  $x, y \in \mathcal{X}$ ,
  - (ii.a) either  $L(x + y) = L(x) \oplus_M L(y)$ ,
  - (ii.b) or  $L(x + y)$  and  $L(x) \oplus_M L(y)$  are LU-incomparable.

It is possible to verify that functions which satisfy (i)-(ii) may not be Fréchet  $gH$ -differentiable, e.g.,  $F(x_1, x_2) = x_1[\alpha - 1, \alpha + 1] + x_2[\beta - 1, \beta + 1]$  and  $x = (0, 1)$  for which the required limit does not exist for  $\|(h_1, h_2)\| \rightarrow 0$  with  $(h_1, h_2) \rightarrow (0^+, 0^-)$ .

A similar observation has been recently advanced in [31], where differentiability of  $IV$ -functions is analyzed in the setting of quasilinear spaces.

Consider also that a homogeneous  $F(x)$  such that  $F(x + y)$  and  $F(x) \oplus_M F(y)$  have the same midpoint value always satisfy conditions (i)-(ii), as, for example,  $F(x_1, x_2) = [\underline{F}(x_1, x_2), \overline{F}(x_1, x_2)]$  with

$$\begin{aligned} \underline{F}(x_1, x_2) &= a_1 x_1 + a_2 x_2 - \alpha \sqrt{b_1 x_1^2 + b_2 x_2^2} \\ \overline{F}(x_1, x_2) &= a_1 x_1 + a_2 x_2 + \alpha \sqrt{b_1 x_1^2 + b_2 x_2^2} \end{aligned}$$

where  $a_i, b_i \in \mathbb{R}$ ,  $b_i > 0$ ,  $i = 1, 2$  and  $\alpha > 0$ .  $\underline{F}(x_1, x_2)$  and  $\overline{F}(x_1, x_2)$  are both positively homogeneous, it is  $F(x + y) \subseteq F(x) \oplus_M F(y)$  and, using midpoint notation, it is  $\hat{f}(x + y) = \hat{f}(x) + \hat{f}(y)$  (i.e.  $\hat{f}$  is linear) and  $\tilde{f}(x + y) \leq \tilde{f}(x) + \tilde{f}(y)$ , but  $\tilde{f}(x) = \alpha \sqrt{b_1 x_1^2 + b_2 x_2^2}$  is not the absolute value of a linear function, i.e., it is not  $gH$ -linear.

For a Fréchet  $gH$ -differentiable  $IV$ -function, the  $gH$ -gradient is defined as a vector of intervals.

**Definition 41.** Let  $F : K \subseteq \mathbb{R}^n \rightarrow I(\mathbb{R})$ ,  $F(x) = (\hat{f}(x); \tilde{f}(x))$  be  $gH$ -differentiable at  $x^{(0)} \in K$ . The  $n$ -dimensional vector of intervals  $W_j = (\hat{w}_j; |\tilde{w}_j|)$ ,  $j = 1, 2, \dots, n$ , is called the  $gH$ -gradient of  $F$  at  $x^{(0)}$  and denoted by

$$\nabla_{gH} F(x^{(0)}) = (W_1, W_2, \dots, W_n).$$

The partial  $gH$ -derivative of  $F$  at  $x^{(0)} \in K$  with respect to the  $j$ -th variable  $x_j$ ,  $j = 1, 2, \dots, n$ , is defined as the directional  $gH$ -derivative in the canonical direction  $e_j$  of  $\mathbb{R}^n$ . If  $F$  is Gateaux  $gH$ -differentiable at  $x^{(0)} \in K$ , then all partial  $gH$ -derivatives of  $F$  exist and, for  $j = 1, 2, \dots, n$ ,

$$\begin{aligned} \frac{\partial_{gH} F(x^{(0)})}{\partial x_j} &= \lim_{t \rightarrow 0} \frac{F(x^{(0)} + te_j) \ominus_{gH} F(x^{(0)})}{t} = \\ &= \left( \hat{w}_j; \left| \tilde{w}_j \right| \right). \end{aligned}$$

The proof of the following fact is immediate:

**Proposition 42.** Consider a Gateaux  $gH$ -differentiable IV-function  $F : K \rightarrow I(\mathbb{R})$ , where  $K$  is an open subset of  $\mathbb{R}^n$ . If  $d \in \mathbb{R}^n$ ,  $\|d\| = 1$ , is any admissible direction at  $x^{(0)} \in K$ , we say that  $F$  has the one-sided directional  $gH$ -derivative at  $x^{(0)}$  in direction  $d$  if the following right limit exists and is an interval:

$$\partial_{gH} F(x^{(0)}; d) = \lim_{t \rightarrow 0^+} \frac{F(x^{(0)} + td) \ominus_{gH} F(x^{(0)})}{t}.$$

In midpoint notation, we will write this interval as  $\left( \widehat{F'_{gH}}(x^{(0)}; d); \widetilde{F'_{gH}}(x^{(0)}; d) \right)$ . If also the left limit for  $t \rightarrow 0^-$  of the function above exists and the two are equal, then  $F$  has the two-sided directional  $gH$ -derivative in direction  $d$  at  $x^{(0)}$ .

**Remark 43.** Recently Roy et al. [36] proposed a Definition of  $gH$ -differentiability for an interval-valued function in several variables similar to that given in expression (24), but replacing  $L_{x^{(0)}}(h)$  for the expression  $h^T \nabla F(x^{(0)})$ , where  $\nabla F(x^{(0)}) = \nabla_{gH} F(x^{(0)})$ . As advanced in the previous section, it means that they replace the requirement of the  $gH$ -linearity in Definition 29 by a form of ‘linearity’ in Minkowski addition and scalar multiplication sense (i.e., an  $M$ -combination of intervals). Then, it is derived the following expression:

$$\lim_{\|h\| \rightarrow 0} \frac{\left\| (F(x^{(0)} + h) \ominus_{gH} F(x^{(0)})) \ominus_{gH} h^T \nabla F(x^{(0)}) \right\|_H}{\|h\|} = 0 \quad (36)$$

Observe that the expression without  $\|\cdot\|_H$  in numerator is equivalent to the previous one. Consider the interval-valued function in two variables  $F : [0, \infty) \times [0, \infty) \subseteq \mathbb{R}^2 \rightarrow I(\mathbb{R})$ , with  $F(x_1, x_2) = [1, 3]x_1 \ominus_M [2, 4]x_2$ . It is not difficult to show that  $F$  is  $gH$ -differentiable, following Definition 29, and calculate the gradient  $\nabla F_{gH}(x^{(0)}) = ((2; 1), (-3; 1))^T$  in midpoint notation, for all  $x^{(0)}$ . Observe that  $(h_1, h_2)^T \nabla F_{gH}(x^{(0)}) = (2h_1 - 3h_2; |h_1| + |h_2|)$  and, on the other hand,  $D_{gH} F(x^{(0)})(h) = (2h_1 - 3h_2; |h_1 + h_2|)$ . These two expressions are different. For instance, if  $x^{(0)} = (5, 5)^T$  and  $h = (1, -1)^T$ , then  $h^T \nabla F_{gH}(x^{(0)}) = (1; 2)$ , and  $D_{gH} F(x^{(0)})(h) = (1; 0)$ . However, the interval-valued function  $F$  is not  $gH$ -differentiable following Definition given in [36]. That is, expression (36) is not fulfilled.

We also recall that, due to complete equivalence of the two representations of compact intervals (end-point and mid-point), we have that any property or expression described in terms of functions  $\hat{f}, \tilde{f}$  can be also described in terms of  $\underline{f}, \overline{f}$  and vice versa.

We have the following immediate connection between Fréchet or Gateaux  $gH$ -differentiability and the existence of (ordinary) directional derivative and partial derivatives of the midpoint and radius functions (see also [45]).

**Theorem 44.** Let  $F = (\hat{f}; \tilde{f})$  be an interval-valued function on an open set  $K \subseteq \mathbb{R}^n$  and assume that  $F$  is Gateaux  $gH$ -differentiable at  $x^{(0)}$ ; if  $\tilde{f}$  has all the (standard) partial derivatives at  $x^{(0)}$  and  $d = (d_1, d_2, \dots, d_n)$  is any fixed direction,  $\|d\| = 1$ , then the (two-sided) directional  $gH$ -derivative of  $F$  at  $x^{(0)}$  is given by

$$\begin{aligned} \partial_{gH} F(x^{(0)}; d) &= \left( \sum_{j=1}^n d_j \frac{\partial \hat{f}(x^{(0)})}{\partial x_j}; \left| \sum_{j=1}^n d_j \frac{\partial \tilde{f}(x^{(0)})}{\partial x_j} \right| \right) \\ &= D_{gH} F(x^{(0)})(d) = F'_{gH}(x^{(0)}; d). \end{aligned} \quad (37)$$

**Theorem 45.** Let  $F = (\hat{f}; \tilde{f})$  be an interval-valued function on an open set  $K \subseteq \mathbb{R}^n$ . If  $F$  is  $gH$ -differentiable at  $x^{(0)}$ , then all the interval-valued partial  $gH$ -derivatives of  $F$  exist and, with the notation used in Definition 38,

$$\begin{aligned} \frac{\partial_{gH} F(x^{(0)})}{\partial x_j} &= F'_{gH}(x^{(0)}; e_j) \\ &= \left( \hat{w}_j; \left| \tilde{w}_j \right| \right) = \left[ \hat{w}_j - \left| \tilde{w}_j \right|, \hat{w}_j + \left| \tilde{w}_j \right| \right], \end{aligned}$$

for  $j = 1, \dots, n$ . Furthermore, all directional  $gH$ -derivatives of  $F$  exist and, for  $\|d\| = 1$ ,

$$\partial_{gH} F(x^{(0)}; d) = \left( \sum_{j=1}^n d_j \hat{w}_j; \left| \sum_{j=1}^n d_j \tilde{w}_j \right| \right).$$

An interesting fact is that any Minkowski combination of  $n$  intervals  $A_j = (\hat{a}_j; \tilde{a}_j)$ ,  $j = 1, \dots, n$ , seems to be Fréchet  $gH$ -differentiable only at points  $x^{(0)} \in \mathbb{R}^n$  devoid of null components, i.e.,  $x_j^{(0)} \neq 0$  (if  $\tilde{a}_j > 0$ ) for  $j = 1, \dots, n$  (for the well known case  $n = 1$  see [45]).

**Proposition 46.** Let  $A_j = (\hat{a}_j; \tilde{a}_j)$ ,  $j = 1, 2, \dots, n$ ,  $n > 1$ , be given intervals and consider the interval-valued function  $F : \mathbb{R}^n \rightarrow I(\mathbb{R})$  defined by  $F(x) = \sum_{j=1}^n x_j A_j = \left( \sum_{j=1}^n x_j \hat{a}_j; \sum_{j=1}^n |x_j| \tilde{a}_j \right)$ . Let  $x^{(0)} = (x_1^{(0)}, x_2^{(0)}, \dots, x_n^{(0)}) \in \mathbb{R}^n$  and assume  $x_j^{(0)} = 0 \implies \tilde{a}_j = 0$  for all  $j$ ; then  $F$  is Fréchet  $gH$ -differentiable at  $x^{(0)}$  and

$$L_{x^{(0)}}(h) = \left( \sum_{j=1}^n h_j \hat{a}_j; \left| \sum_{j=1}^n h_j \tilde{w}_j \right| \right) \text{ with} \quad (38)$$

$$\tilde{w}_j = \begin{cases} \tilde{a}_j & \text{if } x_j^{(0)} \geq 0 \\ -\tilde{a}_j & \text{if } x_j^{(0)} < 0 \end{cases}, \quad j = 1, \dots, n. \quad (39)$$

**Proof.** We have

$$F(x^{(0)} + h) = \left( \sum_{j=1}^n (x_j^{(0)} + h_j) \hat{a}_j; \sum_{j=1}^n |x_j^{(0)} + h_j| \tilde{a}_j \right)$$

so that  $(F(x^{(0)} + h) \ominus_{gH} F(x^{(0)}))$  equals  $\left( \sum_{j=1}^n h_j \hat{a}_j; \left| \sum_{j=1}^n (|x_j^{(0)} + h_j| - |x_j^{(0)}|) \tilde{a}_j \right| \right)$ .

For  $h \neq 0$  and sufficiently small  $\|h\|$ , denoting

$$(\hat{\varphi}(h); \tilde{\varphi}(h)) = (F(x^{(0)} + h) \ominus_{gH} F(x^{(0)})) \ominus_{gH} L_{x^{(0)}}(h)$$

we have  $\hat{\varphi}(h) = 0$  and

$$\tilde{\varphi}(h) = \left| \left( \sum_{j=1}^n (|x_j^{(0)} + h_j| - |x_j^{(0)}|) \tilde{a}_j \right) - \left| \sum_{j=1}^n h_j \tilde{w}_j \right| \right|;$$

on the other hand, for small  $\|h\|$  and each  $j = 1, 2, \dots, n$ , it is

$$(|x_j^{(0)} + h_j| - |x_j^{(0)}|) \tilde{a}_j = \begin{cases} h_j \tilde{a}_j = h_j \tilde{w}_j, & x_j^{(0)} > 0 \\ |h_j| \tilde{a}_j, & x_j^{(0)} = 0 \\ -h_j \tilde{a}_j = h_j \tilde{w}_j, & x_j^{(0)} < 0 \end{cases}$$

and, under the assumption that  $x_j^{(0)} = 0 \implies \tilde{a}_j = 0$ , we conclude that  $\tilde{\varphi}(h) = \left| \sum_{j=1}^n h_j \tilde{w}_j \right| - \left| \sum_{j=1}^n h_j \tilde{w}_j \right| = 0$ . It follows that  $F$  is Fréchet  $gH$ -differentiable with  $L_{x^{(0)}}$  given in equation (38).  $\square$

**Remark 47.** Consider  $n = 2$ ,  $x^{(0)} = (0, 0)$ ,  $\tilde{a}_1, \tilde{a}_2 > 0$ . With  $\tilde{\varphi}(h)$  as in the proof of Proposition 46, the limit  $\lim_{\|h\| \rightarrow 0} \frac{\tilde{\varphi}(h)}{\|h\|}$  cannot exist; it follows that the Minkowski-type combination  $x_1 A_1 \oplus_M x_2 A_2$  is not Fréchet  $gH$ -differentiable at  $(0, 0)$ .

Our final result in this section gives an important relation between Fréchet  $gH$ -derivative (the  $gH$ -differential operator) and the (interval-valued) partial  $gH$ -derivatives.

In general, it is sufficient to consider only the radius component  $\tilde{f}$  of the  $IV$ -function  $F = (\hat{f}; \tilde{f})$ , eventually with  $\tilde{f} = |\tilde{f}_w|$  being in the form of the absolute value of an unsigned function  $\tilde{f}_w$ ; in fact, the midpoint function  $\hat{f}$  is Fréchet differentiable in the ordinary sense and the standard rules apply for it.

From Fréchet  $gH$ -differentiability, we know that the components  $\tilde{w}_j$  in the limit

$$\lim_{\|h\| \rightarrow 0} \frac{|\tilde{f}(x^{(0)} + h) - \tilde{f}(x^{(0)})| - |\tilde{w}^T h|}{\|h\|} = 0 \quad (40)$$

are related to the partial  $gH$ -derivatives  $\frac{\partial_{gH} F(x^{(0)})}{\partial x_j} = A_j = (\hat{a}_j; \tilde{a}_j)$ ,  $\tilde{a}_j \geq 0$ , for all  $j = 1, 2, \dots, n$ , by

$$\lim_{t \rightarrow 0} \left| \frac{\tilde{f}(x^{(0)} + te_j) - \tilde{f}(x^{(0)})}{t} \right| = \tilde{a}_j = |\tilde{w}_j|. \quad (41)$$

The values of  $\tilde{w}_j$  and  $\tilde{a}_j$  can be computed in terms of the following (signed) partial derivatives of  $\tilde{f}$  ( $j = 1, 2, \dots, n$ ):

$$\lim_{t \rightarrow 0^+} \frac{\tilde{f}(x^{(0)} + te_j) - \tilde{f}(x^{(0)})}{t} = m_j^+ \text{ and } \lim_{t \rightarrow 0^-} \frac{\tilde{f}(x^{(0)} + te_j) - \tilde{f}(x^{(0)})}{t} = m_j^-. \quad (42)$$

**Remark 48.** If there is an index  $j \in \{1, 2, \dots, n\}$  with  $m_j^+ \neq m_j^-$  and  $m_j^+ \neq -m_j^-$ , then  $F$  cannot be Fréchet  $gH$ -differentiable (in this case, the limit condition in (40) cannot hold).

**Theorem 49.** Let  $F = (\hat{f}; \tilde{f})$  be an  $IV$ -function on an open set  $K \subseteq \mathbb{R}^n$ . If  $F$  is Fréchet  $gH$ -differentiable at  $x^{(0)} \in K$ , then, according to positions (40), (41) and (42), the  $gH$ -differential  $D_{gH}F(x^{(0)})(h) = \left( \sum_{j=1}^n h_j \hat{w}_j; \left| \sum_{j=1}^n h_j \tilde{w}_j \right| \right)$  is given, for  $j = 1, 2, \dots, n$ , by  $\hat{w}_j = \hat{a}_j$  and

$$\begin{cases} \tilde{w}_j = m_j & \text{if } m_j^+ = m_j^- = m_j, \\ \tilde{w}_j \in \{m_j^+, m_j^-\} & \text{if } m_j^+ = -m_j^- \neq 0. \end{cases}$$

**Proof.** As  $F$  is Fréchet  $gH$ -differentiable,  $\hat{f}$  is differentiable in the ordinary sense, so  $\hat{w}_j = \hat{a}_j$  for all  $j$ . Considering  $\tilde{a}_j = |\tilde{w}_j|$  from position (41) we distinguish the subcases (i):  $m_j^+ = m_j^- = m_j$  and (ii):  $m_j^+ = -m_j^- \neq 0$ ; from (i) we obtain

$$\tilde{a}_j = \begin{cases} \tilde{w}_j & \text{if } m_j \geq 0 \\ -\tilde{w}_j & \text{if } m_j < 0; \end{cases}$$

so that  $\tilde{w}_j = m_j$ . In subcase (ii) we have  $\tilde{a}_j = |m_j^+| = |m_j^-| > 0$  and only one value  $\tilde{w}_j \in \{m_j^-, m_j^+\}$  is the possible choice; on the other hand, only one of the two terms  $m_j^+$  and  $m_j^-$  will be able to give the valid  $\tilde{w}_j$  in the limit (40).  $\square$

In order to obtain the diagonal sign pattern matrix  $U$ , we have the two cases:

Case 1: If  $m_j^+ = m_j^- = m_j$  then  $\tilde{w}_j = m_j$  and

$$\tilde{a}_j = \begin{cases} \tilde{w}_j & \text{if } m_j \geq 0 \\ -\tilde{w}_j & \text{if } m_j < 0; \end{cases}$$

consequently, the  $j$ -th component of the diagonal sign pattern matrix  $U$  is

$$u_j = \begin{cases} +1 & \text{if } m_j \geq 0 \\ -1 & \text{if } m_j < 0. \end{cases}$$

Case 2: If  $m_j^+ = -m_j^- \neq 0$  then we have two possible combinations (subcases) for  $\tilde{w}_j$  and  $u_j$ , say

$$\tilde{w}_j^{(1)} = m_j^+ \text{ and } u_j^{(1)} = \begin{cases} +1 & \text{if } m_j^+ > 0 \\ -1 & \text{if } m_j^+ < 0; \end{cases}$$

or

$$\tilde{w}_j^{(2)} = m_j^- \text{ and } u_j^{(2)} = \begin{cases} +1 & \text{if } \tilde{m}_j^- > 0 \\ -1 & \text{if } \tilde{m}_j^- < 0. \end{cases}$$

Then,  $\tilde{a}_j = |\tilde{w}_j^{(1)}| = |\tilde{w}_j^{(2)}| > 0$  but  $u_j^{(1)}\tilde{a}_j = -u_j^{(2)}\tilde{a}_j$ ; on the other hand, only one of the two terms  $u_j^{(1)}$  and  $u_j^{(2)}$  will be able to generate the correct value  $u_j\tilde{a}_j$  to obtain the valid  $\tilde{w}_j$  in the limit (40).

**Example 50.** Consider  $\tilde{f}(x_1, x_2) = 3|x_1 - 1| + 2|x_2|$ ,  $\hat{f}(x_1, x_2) = 0$  for all  $x_1, x_2$  and let  $x^{(0)} = (1, 0)$ . The signed partial derivatives are  $m_1^- = -3, m_1^+ = 3, m_2^- = -2, m_2^+ = 2$  so that the partial  $gH$ -derivatives of  $F = (\hat{f}; \tilde{f})$  exist, equal to the two intervals  $(0; 3)$  and  $(0; 2)$ . But  $F$  is not Fréchet  $gH$ -differentiable; in particular, the radius component is not abs-differentiable at  $x^{(0)}$ . Indeed, we have  $\tilde{f}(1, 0) = 0$ ,  $|\tilde{f}(x^{(0)} + h) - \tilde{f}(x^{(0)})| = 3|h_1| + 2|h_2|$  and no  $\tilde{w}_1, \tilde{w}_2 \in \mathbb{R}$  can exist such that  $3|h_1| + 2|h_2| - |\tilde{w}_1 h_1 + \tilde{w}_2 h_2| = 0$  for all  $h \in \mathbb{R}^2$  (e.g., if  $h_1 h_2 < 0$ ); as a consequence, the limit of the ratio  $\frac{3|h_1| + 2|h_2| - |\tilde{w}_1 h_1 + \tilde{w}_2 h_2|}{\|h\|}$  cannot be zero for  $(h_1, h_2) \rightarrow (0, 0)$ .

#### 4.2. The $gH$ -tangency function at points of Fréchet $gH$ -differentiability

In this section we define the  $gH$ -tangent function (also called  $gH$ -slope function) of a Fréchet  $gH$ -differentiable function and show its connection with the  $gH$ -gradient.

The basic definition is that of  $gH$ -slope function.

**Definition 51.** If  $F = (\hat{f}; \tilde{f})$  is an  $IV$ -function defined on an open set  $K \subseteq \mathbb{R}^n$  and  $x^{(0)} \in K$  is a point where  $F$  is Fréchet  $gH$ -differentiable with  $D_{gH}F(x^{(0)})(h) = (\hat{l}^T h; |\tilde{w}^T h|)$  the  $gH$ -slope of  $F$  at  $x^{(0)}$  is defined to be the function  $\Phi_{x^{(0)}} : \mathbb{R}^n \rightarrow I(\mathbb{R})$  as follows:

$$\Phi_{x^{(0)}}(x) = \left( \hat{l}^T(x - x^{(0)}); |\tilde{w}^T(x - x^{(0)})| \right) = L(x - x^{(0)}). \quad (43)$$

Its basic meaning is that, for  $x$  near to  $x^{(0)}$ , interval  $\Phi_{x^{(0)}}(y)$  is a “good” approximation of  $F(x) \ominus_{gH} F(x^{(0)})$ , with an error moving quickly to zero when  $x$  moves to  $x^{(0)}$ ; this qualitative interpretation is clearly suggested by condition

$$(F(x) \ominus_{gH} F(x^{(0)})) \ominus_{gH} \Phi_{x^{(0)}}(x) = r(x; x^{(0)}) \|x - x^{(0)}\|, \quad (44)$$

where the  $IV$ -function  $r(y; x)$ ,  $r = (\hat{r}; \tilde{r})$  has  $\lim_{y \rightarrow x} r(y; x) = 0$ .

If we write this in the form  $F(x) \ominus_{gH} F(x^{(0)}) \approx \Phi_{x^{(0)}}(x)$ , then, in terms of Minkowski addition and subtraction, we have the two possibilities

$$F(x) \approx F(x^{(0)}) \oplus_M \Phi_{x^{(0)}}(x), \text{ or } F(x^{(0)}) \approx F(x) \ominus_M \Phi_{x^{(0)}}(x), \quad (45)$$

which give, using midpoint notation,

$$\hat{f}(x) \approx \hat{f}(x^{(0)}) + \hat{l}^T x - \hat{l}^T x^{(0)} \text{ and } \tilde{f}(x) \approx \tilde{f}(x^{(0)}) + |\tilde{w}^T x - \tilde{w}^T x^{(0)}| \quad (46)$$

or

$$\hat{f}(x^{(0)}) \approx \hat{f}(x) - \hat{l}^T x + \hat{l}^T x^{(0)} \text{ and } \tilde{f}(x^{(0)}) \approx \tilde{f}(x) + |\tilde{w}^T x - \tilde{w}^T x^{(0)}|. \quad (47)$$

With respect to the midpoint component we get  $\hat{f}(x) \approx \hat{f}(x^{(0)}) + \hat{l}^T x - \hat{l}^T x^{(0)}$ .

Following the same approach, with respect to the radius component, we have

$$\tilde{f}(x) \approx \tilde{f}(x^{(0)}) + |\tilde{w}^T x - \tilde{w}^T x^{(0)}| \text{ or } \tilde{f}(x) \approx \tilde{f}(x^{(0)}) - |\tilde{w}^T x - \tilde{w}^T x^{(0)}| \quad (48)$$

and the choice is among the two local approximations of  $F(x)$  near  $x^{(0)}$  along direction  $x - x^{(0)}$ , obtained in terms of the two functions below

$$\begin{cases} P_M(x) = \left( f(x^{(0)}) + \hat{l}^T x - \hat{l}^T x^{(0)}; \tilde{f}(x^{(0)}) + |\tilde{w}^T x - \tilde{w}^T x^{(0)}| \right) \text{ or} \\ P_{gH}(x) = \left( f(x^{(0)}) + \hat{l}^T x - \hat{l}^T x^{(0)}; \tilde{f}(x^{(0)}) - |\tilde{w}^T x - \tilde{w}^T x^{(0)}| \right) \end{cases} \quad (49)$$

that have the same value and all the partial  $gH$ -derivatives in common with  $F$  at  $x^{(0)}$ , as is verified immediately, i.e.,:

$$\begin{cases} P_M(x^{(0)}) = P_{gH}(x^{(0)}) = F(x^{(0)}) \text{ and} \\ \frac{\partial_{gH} P_M(x^{(0)})}{\partial x_j} = \frac{\partial_{gH} P_{gH}(x^{(0)})}{\partial x_j} = \frac{\partial_{gH} F(x^{(0)})}{\partial x_j} \text{ for all } j. \end{cases} \quad (50)$$

Now, the choice between  $P_M(x)$  or  $P_{gH}(x)$  requires an additional requirement or knowledge to be fully appropriate, in particular because the existence of all partial  $gH$ -derivatives does not guarantee, in general, that the two approximating functions  $P_M(x)$  or  $P_{gH}(x)$  have the same differential function as  $F(x)$  at  $x^{(0)}$ .

We may then require to search for a possibly stronger tangency condition to be satisfied by the adopted local approximations of  $F$ , similar to the well-known “first-order contact at a point” between two ordinary functions (same value and first derivative).

**Definition 52.** Let  $K \subseteq \mathbb{R}^n$  be an open set and  $F, G : K \subseteq \mathbb{R}^n \rightarrow I(\mathbb{R})$  be two  $IV$ -functions with  $F = (\hat{f}; \tilde{f})$ ,  $\tilde{f} \geq 0$  and  $G = (\hat{g}; \tilde{g})$ ,  $\tilde{g} \geq 0$ . Fix a point  $x^{(0)} \in K$ . We say that  $F$  and  $G$  satisfy the first-order tangency condition at  $x^{(0)}$  if and only if

$$\left\{ \begin{array}{l} G(x^{(0)}) = F(x^{(0)}) = A_0, \text{ and} \\ \lim_{\|h\| \rightarrow 0} \frac{(F(x^{(0)} + h) \ominus_{gH} G(x^{(0)} + h))}{\|h\|} = (0; 0). \end{array} \right. \quad (51)$$

It is clear that (51) establish an equivalence relation between  $IV$ -functions.

The following property will be used to prove the next Proposition.

**Lemma 53.** *If  $F, G$  satisfy the first order tangency condition at  $x^{(0)}$ , then*

$$\lim_{\|h\| \rightarrow 0} \frac{(F(x^{(0)} + h) \ominus_{gH} A_0) \ominus_{gH} (G(x^{(0)} + h) \ominus_{gH} A_0)}{\|h\|} = (0; 0). \quad (52)$$

**Proof.** We apply the following inclusion, valid for any intervals  $A, B, C$ , easy to prove

$$(A \ominus_{gH} C) \ominus_{gH} (B \ominus_{gH} C) \subseteq (A \ominus_{gH} B), \quad (53)$$

applied with  $A = (F(x^{(0)} + h), B = G(x^{(0)} + h)$  and  $C = A_0$ .  $\square$

**Proposition 54.** *Let  $K \subseteq \mathbb{R}^n$  be an open set and  $F, G : K \rightarrow I(\mathbb{R})$  be two  $IV$ -functions,  $gH$ -differentiable at  $x^{(0)} \in K$ . If they satisfy the first order tangency condition at  $x^{(0)}$ , then their  $gH$ -differentials coincide, i.e., for all  $h = x - x^{(0)} \in \mathbb{R}^n$*

$$D_{gH} G(x^{(0)})(h) = D_{gH} F(x^{(0)})(h). \quad (54)$$

**Proof.** Suppose that (51) is true; then, for  $h = te_j$ ,  $j = 1, \dots, n$ , we have  $\|h\| = |t|$  and  $\lim_{t \rightarrow 0} \left| \frac{F(x^{(0)} + te_j) \ominus_{gH} G(x^{(0)} + te_j)}{t} \right| = (0; 0)$ . To see that the differentials of  $F$  and  $G$  coincide, it is sufficient to verify that they have the same signed partial  $gH$ -derivatives, i.e., if  $\frac{\partial_{gH} F(x^{(0)})}{\partial x_j^\pm} = (\hat{a}_j; |m_j^\pm|)$  and  $\frac{\partial_{gH} G(x^{(0)})}{\partial x_j^\pm} = (\hat{b}_j; |n_j^\pm|)$ , i.e.,  $\hat{a}_j = \hat{b}_j$  and  $m_j^\pm = n_j^\pm$  for all  $j$ .

For the midpoint components, the limits exist

$$\lim_{t \rightarrow 0} \frac{\hat{f}(x^{(0)} + te_j) - \hat{a}_0}{t} = \hat{a}_j,$$

$$\lim_{t \rightarrow 0} \frac{\hat{g}(x^{(0)} + te_j) - \hat{a}_0}{t} = \hat{g}_j, \text{ and}$$

$$\lim_{t \rightarrow 0} \frac{\hat{f}(x^{(0)} + te_j) - \hat{g}(x^{(0)} + te_j)}{t} = 0; \text{ then}$$

$$\hat{a}_j - \hat{b}_j = \lim_{t \rightarrow 0} \left( \frac{\hat{f}(x^{(0)} + te_j) - \hat{a}_0}{t} - \frac{\hat{g}(x^{(0)} + te_j) - \hat{a}_0}{t} \right) = 0 \text{ and } \hat{a}_j = \hat{b}_j \text{ for all } j.$$

For the radius components, we have that for all  $\epsilon > 0$  there exists  $\delta > 0$  such that:

$$0 < t < \delta \Rightarrow m_j^+ - \epsilon < \frac{\tilde{f}(x^{(0)} + te_j) - \tilde{a}_0}{t} < m_j^+ + \epsilon,$$

$$-\delta < t < 0 \Rightarrow m_j^- - \epsilon < \frac{\tilde{f}(x^{(0)} + te_j) - \tilde{a}_0}{t} < m_j^- + \epsilon,$$

$$0 < t < \delta \Rightarrow n_j^+ - \epsilon < \frac{\tilde{g}(x^{(0)} + te_j) - \tilde{a}_0}{t} < n_j^+ + \epsilon,$$

$$-\delta < t < 0 \Rightarrow n_j^- - \epsilon < \frac{\tilde{g}(x^{(0)} + te_j) - \tilde{a}_0}{t} < n_j^- + \epsilon \text{ and}$$

$$0 < t < \delta \Rightarrow m_j^- - \epsilon \leq \frac{\tilde{f}(x^{(0)} + te_j) - \tilde{g}(x^{(0)} + te_j)}{t} < \epsilon. \text{ Then}$$

$$0 = \lim_{t \rightarrow 0} \left( \frac{\tilde{f}(x^{(0)} + te_j) - \tilde{a}_0}{t} - \frac{\tilde{g}(x^{(0)} + te_j) - \tilde{a}_0}{t} \right) \in \{m_j^+ - n_j^+, m_j^- - n_j^-\}, \text{ so that } m_j^+ = n_j^+, m_j^- = n_j^- \text{ and } \tilde{f}, \tilde{g} \text{ have the same absolute differential function. } \square$$

The following Theorem is a key property of the Fréchet  $gH$ -differential function in terms of  $gH$ -tangency; here, interval  $A$  is said to be *inclusion-minimal for a property "P"* if and only if, for any other interval  $B$  with the same property "P", we have  $A \subseteq B$ .

**Theorem 55.** *Given an  $IV$ -function  $F : K \subseteq \mathbb{R}^n \rightarrow I(\mathbb{R})$ ,  $F = (\hat{f}; |\tilde{f}_w|)$ , Fréchet  $gH$ -differentiable at  $x^{(0)} \in K$  with  $gH$ -differential  $L_{x^{(0)}}(h) = (\hat{l}^T h; |\tilde{w}^T h|)$ , the function  $P^* : \mathbb{R}^n \rightarrow I(\mathbb{R})$ , defined for all  $x \in \mathbb{R}^n$  by*

$$P^*(x) = \left( \hat{f}(x^{(0)}) + \hat{l}^T(x - x^{(0)}); \left| \tilde{f}_w(x^{(0)}) + \tilde{w}^T(x - x^{(0)}) \right| \right) \quad (55)$$

is inclusion-minimal over all functions that satisfy the tangency condition to  $F$  at  $x^{(0)}$ ; furthermore, for points  $x \in \mathbb{R}^n$  sufficiently near to  $x^{(0)}$ , it is

$$P^*(x) \ominus_{gH} F(x^{(0)}) \approx L_{x^{(0)}}(x - x^{(0)}). \quad (56)$$

**Proof.** Denote for simplicity  $F(x^{(0)}) = F_0 = (\hat{f}_0; \tilde{f}_0)$ ,  $\tilde{f}_0 \geq 0$ . Consider the  $IV$ -function  $P^*(x) = (\hat{p}^*(x); \tilde{p}^*(x))$  with midpoint and radius components, respectively,

$$\begin{aligned} \hat{p}^*(x) &= \hat{p}_0^* + \hat{p}_1^*(x_1 - x_1^{(0)}) + \hat{p}_2^*(x_2 - x_2^{(0)}) + \dots + \hat{p}_n^*(x_n - x_n^{(0)}), \\ \tilde{p}^*(x) &= \left| \tilde{p}_0^* + \tilde{p}_1^*(x_1 - x_1^{(0)}) + \tilde{p}_2^*(x_2 - x_2^{(0)}) + \dots + \tilde{p}_n^*(x_n - x_n^{(0)}) \right|. \end{aligned}$$

On the other hand, from  $L_{x^{(0)}}(x) = (\hat{l}(x); |\tilde{w}(x)|)$  we have

$$\begin{aligned} \hat{l}(x - x^{(0)}) &= \hat{l}_1(x_1 - x_1^{(0)}) + \hat{l}_2(x_2 - x_2^{(0)}) + \dots + \hat{l}_n(x_n - x_n^{(0)}), \\ \tilde{w}(x - x^{(0)}) &= \tilde{w}_1(x_1 - x_1^{(0)}) + \tilde{w}_2(x_2 - x_2^{(0)}) + \dots + \tilde{w}_n(x_n - x_n^{(0)}) \end{aligned}$$

and the  $2n+2$  coefficients  $\hat{p}_j^*$  and  $\tilde{p}_j^*$  for  $j = 0, 1, 2, \dots, n$  are determined from the  $2n+2$  known values  $\hat{f}_0, \tilde{f}_0$  and  $\hat{l}_j, \tilde{w}_j$  for  $j = 1, 2, \dots, n$  such that, with our notation,

$$P^*(x) \ominus_{gH} F_0 = \left( \hat{p}^*(x) - \hat{f}_0; \left| \tilde{p}^*(x) - \tilde{f}_0 \right| \right);$$

then, if we choose

$$\begin{cases} \hat{p}_0^* = \hat{f}_0, & \tilde{p}_0^* = \tilde{f}_0 \geq 0 \\ \hat{p}_j^* = \hat{l}_j, & \tilde{p}_j^* = \tilde{w}_j, \quad \text{for } j = 1, 2, \dots, n \end{cases} \quad (57)$$

we get the  $IV$ -function in (55), which clearly satisfies  $P^*(x^{(0)}) = F_0$  and is Fréchet  $gH$ -differentiable at  $x^{(0)}$  with  $gH$ -differential function  $L_{x^{(0)}}(h)$  for all  $h \in \mathbb{R}^n$ . Finally, to verify (56) with the values in (57), we have  $\hat{p}^*(x) - \hat{f}_0 = \hat{l}(x)$  and  $\tilde{p}^*(x) - \tilde{f}_0 = \left| \tilde{f}_0 - \tilde{w}(x) \right|$  or  $\left| \tilde{f}_0 + \tilde{w}(x) \right|$  with  $\hat{l}(0) = \tilde{w}(0) = 0$ . If  $\tilde{f}_0 = 0$ , then (56) is obviously satisfied; if  $\tilde{f}_0 > 0$ , then for  $x$  sufficiently close to  $x^{(0)}$  it is  $\tilde{f}_0 \pm \tilde{w}(x - x^{(0)}) > 0$  so that  $\left| \tilde{p}^*(x - x^{(0)}) - \tilde{f}_0 \right| = \left| \tilde{f}_0 \pm \tilde{w}(x - x^{(0)}) - \tilde{f}_0 \right| = \left| \tilde{w}(x - x^{(0)}) \right|$  and again (56) is satisfied.  $\square$

#### Example 56. ( $n=3$ )

Consider  $F = (\hat{f}; \tilde{f})$  with  $\hat{f}, \tilde{f} : \mathbb{R}^3 \rightarrow \mathbb{R}$  given by  $\hat{f}(x_1, x_2, x_3) = x_1^2 + x_1 + 2x_2^2(1 - x_3) - x_2 + x_3 + 1$  and  $\tilde{f}(x_1, x_2, x_3) = (x_1^2 - x_2^2 + x_1 - x_2 + x_3 + 1)^2$ ; take  $x^{(0)} = (0, 0, 0)$  so that  $F(x^{(0)}) = (1; 1) = [0, 2]$ .

The two functions  $\hat{f}, \tilde{f}$  are differentiable at all points so that  $F$  is Fréchet  $gH$ -differentiable; at  $x^{(0)} = 0$  their gradients are  $\nabla \hat{f}(0) = (1, -1, 1)^T$ ,  $\nabla \tilde{f}(0) = (2, -2, 2)^T$ .

The  $gH$ -differential function at the origin is  $L(x) = (x_1 - x_2 + x_3; |2x_1 - 2x_2 + 2x_3|)$  with  $\hat{l} = (1, -1, 1)^T$ ,  $\tilde{w} = (2, -2, 2)^T$ . The interval-valued  $gH$ -tangency function is then  $P^* = (\hat{p}^*; \tilde{p}^*)$  with  $\hat{p}^*(x) = 1 + x_1 - x_2 + x_3$  and  $\tilde{p}^*(x) = |1 + 2x_1 - 2x_2 + 2x_3|$ .

A graphical representation of the midpoint and radius component functions requires four dimensions, but we can picture, e.g., the set of intervals that are the images via  $F$  of box  $[-5/4, 5/4]^3 \subset \mathbb{R}^3$ , centered at 0, in the  $(\hat{z}; \tilde{z})$  half plane (with  $\tilde{z} \geq 0$ ); the same can be done for the intervals that are images via the  $gH$ -tangency function  $P^*$  of a neighborhood of 0 (say  $[-0.2, 0.2]^3$ ).

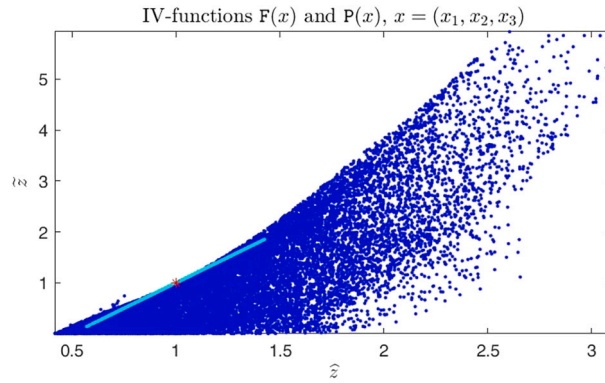
The resulting midpoint-radius images are pictured in Fig. 2. In Fig. 3 (top alignment) the restrictions of functions  $F(x), P^*(x)$  along the three canonical directions  $x_1 e_1 = (x_1, 0, 0)$ ,  $x_2 e_2 = (0, x_2, 0)$ ,  $x_3 e_3 = (0, 0, x_3)$  are pictured in the midpoint-radius halfplane; they are simple unidimensional  $IV$  functions. In the bottom three pictures, analogous restrictions of  $F(x), P^*(x)$  on the three subspaces described by points  $x = (0, x_2, x_3)$ ,  $x = (x_1, 0, x_3)$ ,  $x = (x_1, x_2, 0)$  are pictured; they are essentially functions of two variables.

#### 4.3. Convexity concepts for $IV$ -functions

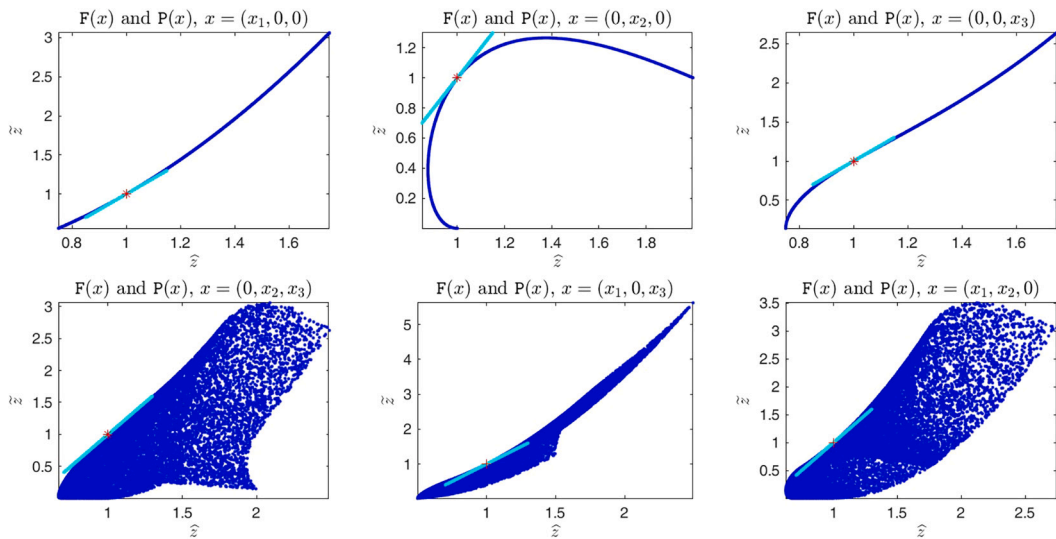
There exist several definitions of convexity for interval-valued functions, based on the LU-order or the inclusion order, in combination with Minkowski addition (see, e.g., [49]).

More generally, the convexity/concavity of an interval-valued function  $F(x) = [\underline{f}(x), \overline{f}(x)]$  on a convex set  $\mathbb{X} \subseteq \mathbb{R}^n$  is defined by requiring that  $F(\alpha x + \beta y)$  is comparable with  $\alpha F(x) \oplus_M \beta F(y)$  for all  $x, y \in \mathbb{X}$  and all  $\alpha, \beta \in [0, 1]$ ,  $\alpha + \beta = 1$ .

**Definition 57.** Given a function  $F(x) = [\underline{f}(x), \overline{f}(x)]$  on a convex set  $\mathbb{X} \subseteq \mathbb{R}^n$ , we say that



**Fig. 2.** Midpoint graphical representation (in the half-plane  $(\hat{z}; \bar{z}), \bar{z} \geq 0$ ) of  $F(x), x \in [-5/4, 5/4]^3$  (in blue color) and intervals  $P^*(x), x \in [-0.2, 0.2]^3$  (cyan colored) of the  $gH$ -tangency function at the origin. The intervals are sampled randomly with  $m = 3000$  points, uniform on the box. Remark that interval  $F(0) = (1; 1)$  has the endpoint representation  $[0, 2]$  (red colored with marker +).



**Fig. 3.** Midpoint graphical representation of restriction of  $F(x)$  (in blue color) and  $P^*(x)$  (cyan colored) of the  $gH$ -tangency function at the origin. *Top three pictures:* restrictions of  $F$  along the three canonical directions, i.e.,  $F(re_j), j : 1, 2, 3, r \in [-5/4, 5/4]$ . *Bottom three pictures:* restrictions of  $F$  on the three 2-dimensional subspaces  $(x_1, x_2, 0), (x_1, 0, x_2), (0, x_2, x_3), x_j \in [5/4, 5/4]$ . The intervals are sampled uniformly with  $m = 2000$  points. Interval  $F(0) = (1; 1) = [0, 2]$  is red colored with marker +.

(i) (lattice  $\leq$ -convexity):  $F$  is  $\leq$ -convex if and only if

$$F(\alpha x + \beta y) \leq \alpha F(x) \oplus_M \beta F(y) \quad (58)$$

for all  $x, y \in \mathbb{X}$  and all  $\alpha, \beta \in [0, 1], \alpha + \beta = 1$ ,

(ii) (strict  $\leq$ -convexity):  $F$  is  $\leq$ -convex if and only if

$$F(\alpha x + \beta y) \leq \alpha F(x) \oplus_M \beta F(y) \quad (59)$$

for all  $x \neq y \in \mathbb{X}$  and all  $\alpha, \beta \in [0, 1], \alpha + \beta = 1$ ,

(iii) (strong  $<$ -convexity):  $F$  is  $<$ -convex if and only if

$$F(\alpha x + \beta y) < \alpha F(x) \oplus_M \beta F(y) \quad (60)$$

for all  $x \neq y \in \mathbb{X}$  and all  $\alpha, \beta \in ]0, 1[, \alpha + \beta = 1$ .

Lattice, strict and strong  $\leq$ -concavity are defined by changing the direction of the partial order.

**Definition 58.** By considering the inclusion-based partial order, we can define conditions for lattice  $\subseteq$ -convexity, strict  $\subseteq$ -convexity and strong  $\subseteq$ -convexity, similar to the three cases above, by changing symbol  $<$  with  $\subset$ . The three forms of concavity are obtained analogously.

**Proposition 59.** Let  $F = (\hat{f}; \tilde{f})$  with  $\tilde{f} \geq 0$ ,  $\hat{f}, \tilde{f} : \mathbb{X} \subseteq \mathbb{R}^n \longrightarrow \mathbb{R}$  be continuous on convex set  $\mathbb{X}$ ; then

1)  $F$  is  $\leq$ -convex if and only if, for all  $x, y \in \mathbb{X}$  and all  $\alpha, \beta \in [0, 1]$ ,  $\alpha + \beta = 1$ ,

$$|\alpha \tilde{f}(x) + \beta \tilde{f}(y) - \tilde{f}(\alpha x + \beta y)| \leq \alpha \hat{f}(x) + \beta \hat{f}(y) - \hat{f}(\alpha x + \beta y) \quad (61)$$

and, consequently,  $\hat{f}, f = \hat{f} - \tilde{f}$  and  $\bar{f} = \hat{f} + \tilde{f}$  are convex;

2)  $F$  is  $\subseteq$ -convex if and only if

$$|\alpha \hat{f}(x) + \beta \hat{f}(y) - \hat{f}(\alpha x + \beta y)| \leq \alpha \tilde{f}(x) + \beta \tilde{f}(y) - \tilde{f}(\alpha x + \beta y) \quad (62)$$

and, consequently,  $\tilde{f}, \bar{f} = \tilde{f} + \hat{f}$  and  $-f = \tilde{f} - \hat{f}$  are convex.

**Proof.** Define, for simplicity, the intervals

$$A = (\hat{a}; \tilde{a}) = F(\alpha x + \beta y) = (\hat{f}(\alpha x + \beta y); \tilde{f}(\alpha x + \beta y)) \text{ and}$$

$$B = (\hat{b}; \tilde{b}) = \alpha F(x) \oplus_M \beta F(y) = (\hat{f}(\alpha x) + \hat{f}(\beta y); \tilde{f}(\alpha x) + \tilde{f}(\beta y)).$$

The  $\leq$ -convexity condition is  $A \leq B$ , equivalent to the three conditions  $\hat{a} \leq \hat{b}$ ,  $\tilde{b} - \tilde{a} \leq \tilde{b} - \hat{a}$  and  $\tilde{b} - \tilde{a} \geq \hat{a} - \hat{b}$ ; the second and third can be grouped into  $|\tilde{b} - \tilde{a}| \leq \hat{b} - \hat{a}$  and the required condition in 1) is obtained; they also display that  $\hat{f}$  is convex,  $\hat{a} - \tilde{a} \leq \hat{b} - \tilde{b}$  (i.e.,  $f$  is convex) and  $\hat{a} + \tilde{a} \leq \hat{b} + \tilde{b}$  (i.e.,  $\bar{f}$  is convex).

Similarly,  $\subseteq$ -convexity condition is  $A \subseteq B$ , equivalent to the three conditions  $\tilde{a} \leq \tilde{b}$ ,  $\hat{a} - \tilde{a} \geq \hat{b} - \tilde{b}$  and  $\hat{a} + \tilde{a} \leq \hat{b} + \tilde{b}$ ; the condition required in 2) is obtained by grouping the second and third into  $|\hat{b} - \hat{a}| \leq \tilde{b} - \tilde{a}$ ; we also have that  $\tilde{f}$  is convex,  $-\hat{a} + \tilde{a} \leq -\hat{b} + \tilde{b}$  (i.e.,  $-f$  is convex) and  $\hat{a} + \tilde{a} \leq \hat{b} + \tilde{b}$  (i.e.,  $\bar{f}$  is convex).  $\square$

In particular, for a continuous  $IV$ -function  $F = (\hat{f}; \tilde{f})$  with  $\tilde{f} \geq 0$ , a necessary and sufficient condition for  $\leq$ -convexity (locally or globally) is that  $\tilde{f}$  is the difference of convex functions (DC function, [21]); and a necessary and sufficient condition for  $\subseteq$ -convexity (locally or globally) is that  $\hat{f}$  is a DC function. This is an important difference between the two types of partial  $\leq$  and  $\subseteq$  orders.

The mathematical concepts related to convexity, including the analysis of extremal points (maxima or minima) with respect to the two partial orders  $\leq$  and  $\subseteq$ , by focusing on basic definitions, properties and conditions for their characterizations (global/local max/min points, dominating and efficient regions) in relation with convexity and  $gH$ -derivatives is a fundamental topic of research.

The case of interval valued functions of a single variable has been extensively analyzed in two companion papers published in 2019 (see [47], [48]), where a very general family of partial orders has been considered (as proposed in [19]) and lattice-type, strict or strong dominances are studied. The case of  $IV$ -functions of multiple variables in the setting of  $gH$ -differentiability is being the subject of our current research but is not the scope of this paper.

Remark that, in many cases, we can use directional (Gateaux)  $gH$ -derivatives to approximate  $IV$ -functions locally or to define partial or directional tangency conditions, as is done e.g. in some examples of next section.

#### 4.4. Some interval-valued examples

In the first example, both  $\hat{f}$  and  $\tilde{f}$  are differentiable, at the reference point; the Fréchet  $gH$ -differentiability is immediate.

**Example 60.** Consider  $F = (\hat{f}; \tilde{f})$  with  $\hat{f}(x_1, x_2) = x_1^4 + x_1 x_2 + x_2$ ,  $\tilde{f}(x_1, x_2) = 1/2 + x^2 + (x_1 - x_2 + x_1 x_2)^2$  for all  $x \in [-1, 1]^2$  and let  $x^{(0)} = (0, 0)$ . Both  $\hat{f}, \tilde{f}$  are differentiable with gradients at  $x^{(0)}$  given by  $(0, 1)^T$  and  $(0, 0)^T$ , respectively;  $F$  is Fréchet  $IV$   $gH$ -differentiable with  $\hat{I}^T h = 1 - h_2$  and  $\tilde{w}^T h = 0$ . See Fig. 4.

**Example 61.** Consider  $F$  with  $\hat{f}(x_1, x_2) = \frac{1}{2}((1 + x_1)^2(1 - x_2^2) + (1 - x_1^2)(1 + x_2)^2)$ , and  $\tilde{f}(x_1, x_2) = \frac{1}{2}(|x_1| + |x_2|) + |x_1 + x_2|$  for all  $x \in [-1.2, 1.2]^2$  and let  $x^{(0)} = (0, 0)$ . Function  $\hat{f}$  is differentiable at the origin with gradient  $(1, 1/2)$ ; function  $\tilde{f}$  is not differentiable at the origin but it is abs-differentiable with left/right partial derivatives  $m_1^- = -1, m_1^+ = 1, m_2^- = -1, m_2^+ = -1$  and the abs-gradient is  $(1, 1)$ .  $F$  is Fréchet  $IV$   $gH$ -differentiable with  $\hat{I}^T h = h_1 + h_2/2$  and  $\tilde{w}^T h = 0$ . See Fig. 5.

**Example 62.** In this example, we consider  $F = (\hat{f}; \tilde{f})$  with

$$\begin{aligned} \hat{f}(x_1, x_2) &= 1 + 2(x_1 - 1)^2 + (x_2 - 1)^2, \text{ and} \\ \tilde{f}(x_1, x_2) &= \begin{cases} 1 + (x_1 - 1)^2 + x_2^2 & \text{if } x_1 < 1, x_2 \in \mathbb{R} \\ |x_1 - 1| + x_1^2 + x_2^2 & \text{if } x_1 \geq 1, x_2 \in \mathbb{R}. \end{cases} \end{aligned}$$

We limit the domain to  $x \in [0, 2] \times [-2, 2]$ . At point  $x^{(0)} = (1, 0)$ ,  $\hat{f}$  is differentiable with gradient  $(0, -2)$  and differential function  $(h_1, h_2) \rightarrow -2h_2$ ; the left and right partial derivatives of  $\tilde{f}$  at  $x^{(0)}$  w.r.t.  $x_1$  are, respectively, 0 and  $-2$  and w.r.t.  $x_2$  are both 0, so  $\tilde{f}$

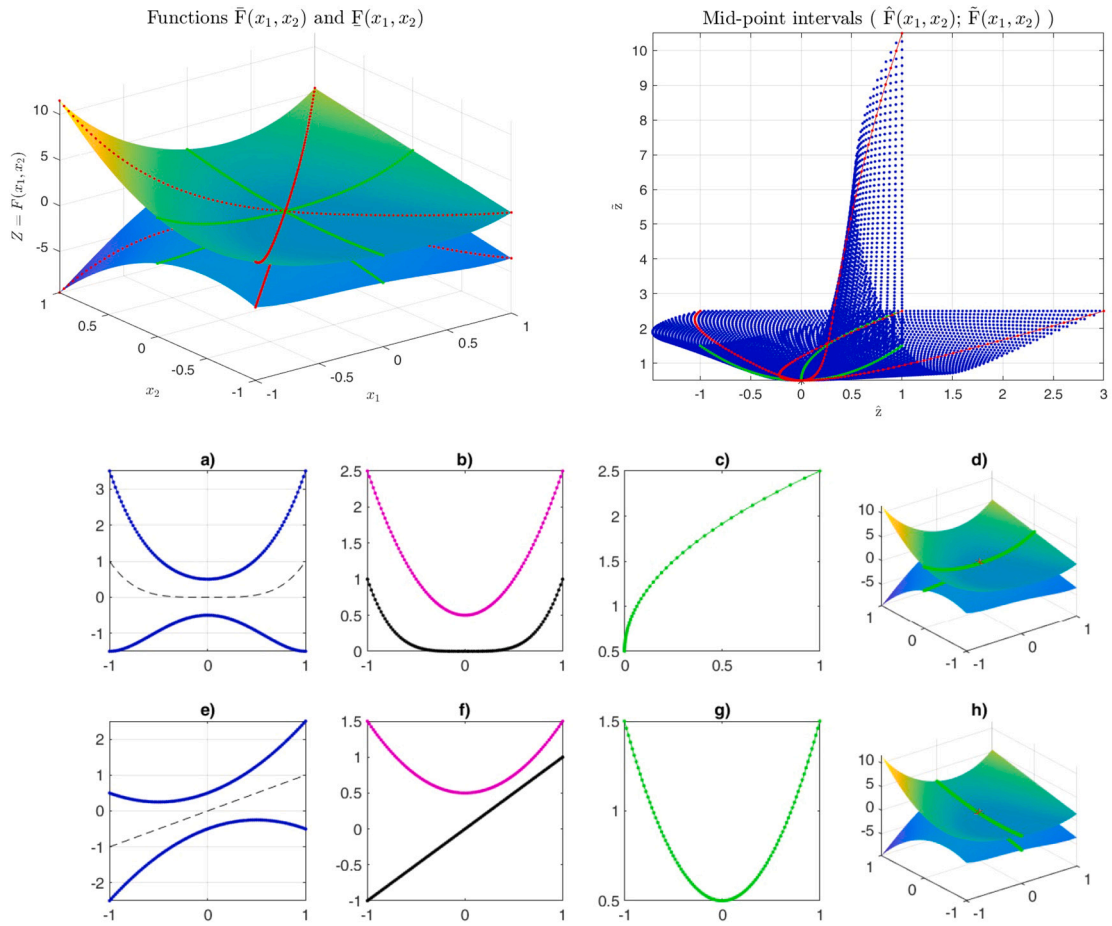


Fig. 4. This figure refers to the  $IV$ -function  $F(x_1, x_2)$  described in Example 60, with  $(x_1, x_2) \in [-1, 1]^2$ . Top-left: Lower and upper surfaces corresponding to the end-points of each interval  $F(x_1, x_2)$ . Top-right: each point  $(\hat{z} = \hat{F}(x_1, x_2); \tilde{z} = \tilde{F}(x_1, x_2))$  represents an interval in mid-point form, the pictured lines correspond to intervals along the two canonical directions  $e_1 = (1, 0)$ ,  $e_2 = (0, 1)$  (green lines) and along the two diagonal directions  $d_1 = (1, 1)$ ,  $d_2 = (-1, 1)$ . The restrictions of  $F$  along the two canonical directions are pictured as end-point, pictures a) and e), and as mid-point curves, pictures c) and d), so resulting as unidimensional  $IV$ -functions.

is not abs-differentiable and  $F$  is not Fréchet  $gH$ -differentiable. This is well seen in Fig. 6; the restrictions of  $\hat{f}, \tilde{f}$  along the diagonal directions  $(1+t, 1+t)$ , in picture b), and  $(-1+t, 1+t)$  in picture f), for  $t \in [-1, 1]$  so that  $x^{(0)}$  corresponds to  $t=0$ , show that  $\hat{f}$  is a differentiable parabola (black curves), while left and right derivatives of  $\tilde{f}(1+t, 1+t)$  w.r.t.  $t$  have different absolute value.

## 5. An outline of extending Fréchet $gH$ -differentiability to interval-valued vector functions and to higher order

The  $gH$ -linearity concept introduced in this paper can be extended to define higher order Fréchet  $gH$ -differentiability and the  $gH$ -Jacobian matrix for a vector  $IV$ -function.

### 5.1. The $gH$ -Jacobian of a vector interval-valued function

Let  $(I(\mathbb{R}))^m$  denote the space of  $m$ -vectors (columns) of intervals; we define on it the interval operations component-wise:

$$tV = (tV_1, tV_2, \dots, tV_m)^T \text{ for } t \in \mathbb{R},$$

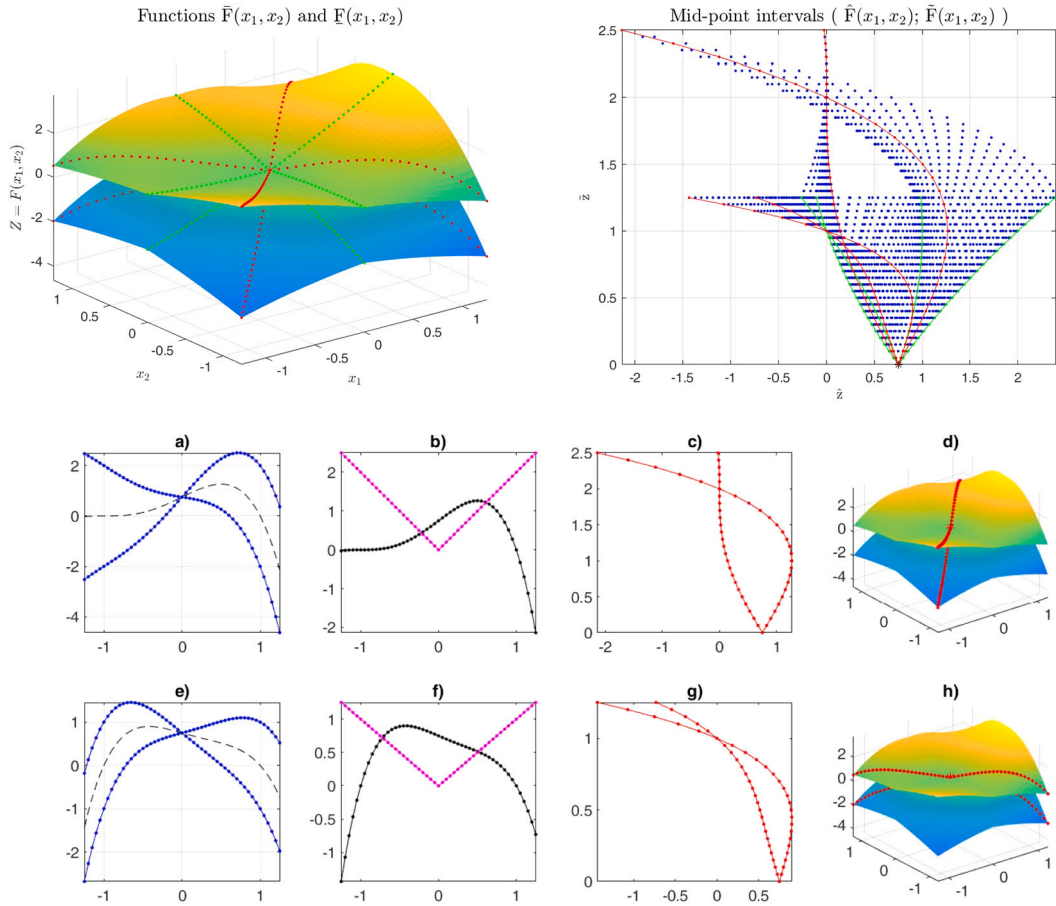
$$V \oplus_M W = (V_1 \oplus_M W_1, V_2 \oplus_M W_2, \dots, V_m \oplus_M W_m)^T$$

and similarly for  $V \ominus_{gH} W, V \otimes_{gH} W$  in terms of components  $V_i \ominus_{gH} W_i, V_i \otimes_{gH} W_i, i = 1, \dots, m$ , respectively.

A  $gH$ -linear combination of  $m$ -vectors of intervals  $A_j = (A_{1,j}, A_{2,j}, \dots, A_{m,j})^T \in (I(\mathbb{R}))^m, j = 1, 2, \dots, n$  with real coefficients  $x_1, x_2, \dots, x_n$  is defined accordingly as the vector  $B = (B_1, B_2, \dots, B_m)^T \in (I(\mathbb{R}))^m$  with components  $B_i \in I(\mathbb{R})$

$$B_i = \bigoplus_{j=1}^n x_j A_{i,j} = \left( \sum_{j=1}^n x_j \hat{a}_{i,j}; \left| \sum_{j=1}^n x_j \tilde{a}_{i,j} \right| \right) \quad (63)$$

(here, each  $A_{i,j} = (\hat{a}_{i,j}, \tilde{a}_{i,j})$  is an interval with real  $\hat{a}_{i,j}, \tilde{a}_{i,j}$  and  $\tilde{a}_{i,j} \geq 0$ ).



**Fig. 5.** This figure refers to the  $IV$ -function  $F(x_1, x_2)$  described in Example 61, with  $(x_1, x_2) \in [-1.2, 1.2]^2$ . *Top-left:* Lower and upper surfaces corresponding to the end-points of each interval  $F(x_1, x_2)$ . *Top-right:* each point  $(\hat{z} = \hat{F}(x_1, x_2); \tilde{z} = \tilde{F}(x_1, x_2))$  represents an interval in mid-point form, the pictured lines correspond to intervals along the two canonical directions  $e_1 = (1, 0)$ ,  $e_2 = (0, 1)$  (green lines) and along the two diagonal directions  $d_1 = (1, 1)$ ,  $d_2 = (-1, 1)$ . The restrictions of  $F$  along the two diagonal directions are pictured as end-point, pictures a) and e), and as mid-point curves, pictures c) and d), so resulting as unidimensional  $IV$ -functions.

A  $m$ -vector interval-valued function of  $n$  variables with values in  $(I(\mathbb{R}))^m$  is then any function  $F : K \subseteq \mathbb{R}^n \rightarrow (I(\mathbb{R}))^m$  with  $F(x) = (F_1(x), F_2(x), \dots, F_m(x))^T$  and  $F_i(x) = (\hat{f}_i(x); \tilde{f}_i(x)) \in I(\mathbb{R})$ ,  $F_i : \mathbb{R}^n \rightarrow I(\mathbb{R})$  for all  $i = 1, 2, \dots, m$ .

If all functions  $F_i(x)$  are Fréchet  $gH$ -differentiable at  $x \in K$  with  $gH$ -derivative  $D_{gH}F_i(x)$  and gradient vectors  $\nabla_{gH}F_i(x) = (G_{i,1}, G_{i,2}, \dots, G_{i,n})^T$  for all  $i = 1, 2, \dots, m$  and partial  $gH$ -derivatives  $\frac{\partial_{gH}F_i(x)}{\partial x_j} = G_{i,j}$ , the matrix of intervals  $J_{gH}F(x)$  of order  $m \times n$  and elements  $(J_{gH}F(x))_{i,j} = G_{i,j}$  is the  $gH$ -Jacobian matrix of  $F$  at  $x$ .

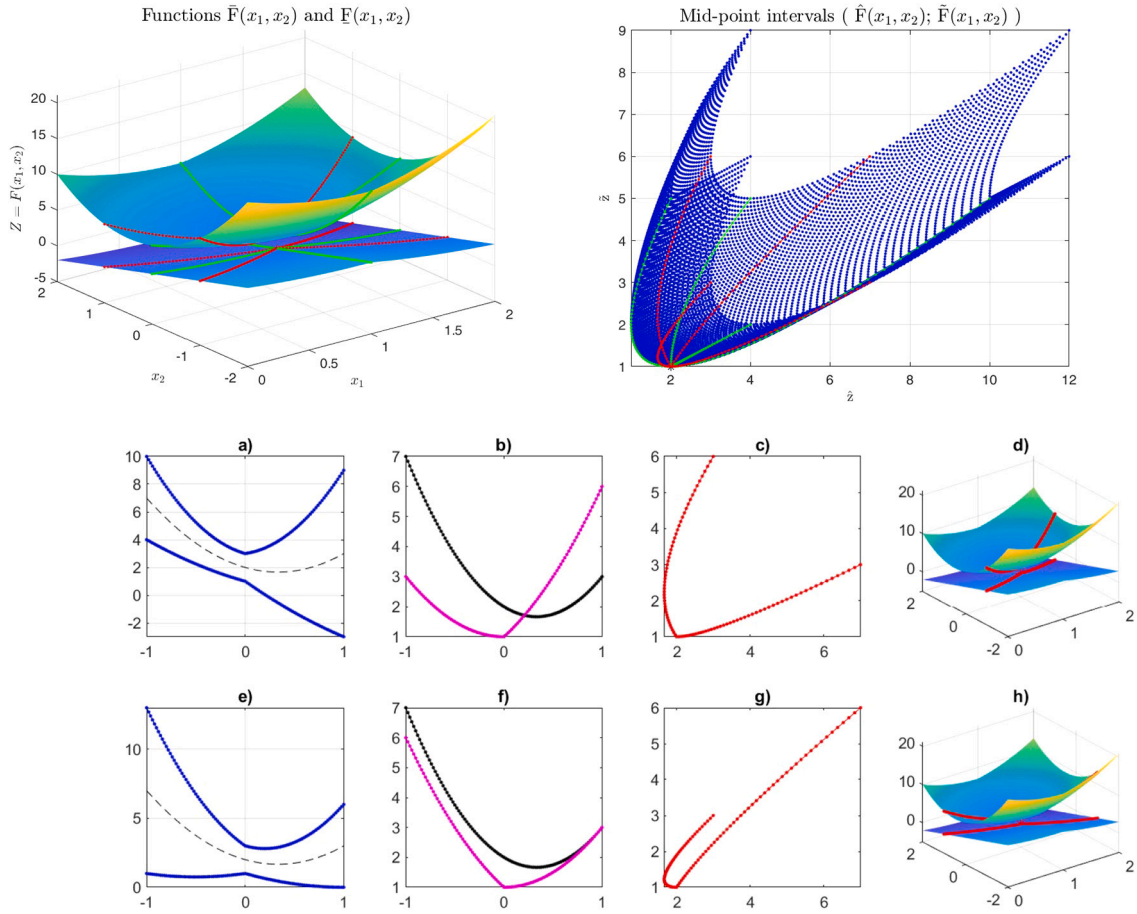
Then, the  $i$ -th row of the  $gH$ -Jacobian matrix contains the components of the  $gH$ -gradient of function  $F_i(x)$ , i.e., the partial interval-valued  $gH$ -derivatives of  $F_i(x)$  with respect to the variables  $x_1, x_2, \dots, x_n$ .

Each interval-valued component of the  $gH$ -Jacobian matrix is interval  $G_{i,j} = (\hat{g}_{i,j}; \tilde{g}_{i,j})$ ,  $\tilde{g}_{i,j} \geq 0$  and we can decompose  $J_{gH}F(x)$  into two matrices  $\hat{J}_{gH}F(x)$  and  $\tilde{J}_{gH}F(x)$ , both of order  $m \times n$ , by forming the matrix  $\hat{J}_{gH}F(x)$  with the midpoint values  $\hat{g}_{i,j}$  and the matrix  $\tilde{J}_{gH}F(x)$  with the radius (half-length) values  $\tilde{g}_{i,j}$ . For the  $gH$ -Jacobian of  $F$  at  $x$ , we will then adopt the following interval notation

$$J_{gH}F(x) = (\hat{J}_{gH}F(x); \tilde{J}_{gH}F(x)), \quad (64)$$

where the two matrices  $\hat{J}_{gH}F(x)$  and  $\tilde{J}_{gH}F(x)$  are standard real matrices.

The  $gH$ -linear vector-function  $L : \mathbb{R}^n \rightarrow I(\mathbb{R})^m$  satisfying the definition of Fréchet  $gH$ -differentiability for each component  $F_i(x)$ , i.e.,  $L_{i,j} = (\hat{l}_{i,j}; \tilde{w}_{i,j})$  with  $\hat{l}_{i,j}, \tilde{w}_{i,j} \in \mathbb{R}$ , is obtained in terms of two standard matrices of order  $(m, n)$ , say  $\hat{L} = [\hat{l}_{i,j}]$ ,  $\tilde{W} = [\tilde{w}_{i,j}]$  by defining  $L(x) = (\hat{L}x; |\tilde{W}x|)$ , where the absolute value of a  $m$ -vector  $y = (y_1, \dots, y_m)^T$  is intended to be obtained component-wise as  $|y| = (|y_1|, \dots, |y_m|)^T$ .



**Fig. 6.** This figure refers to the  $IV$ -function  $F(x_1, x_2)$  described in Example 62, with  $(x_1, x_2) \in [0, 2] \times [-2, 2]$ . *Top-left:* Lower and upper surfaces corresponding to the end-points of each interval  $F(x_1, x_2)$ . *Top-right:* each point  $(\hat{z} = \hat{F}(x_1, x_2); \tilde{z} = \tilde{F}(x_1, x_2))$  represents an interval in mid-point form, the pictured lines correspond to intervals along the two canonical directions  $e_1 = (1, 0), e_2 = (0, 1)$  (green lines) and along the two diagonal directions  $d_1 = (1, 1), d_2 = (-1, 1)$ . The same restrictions of  $F$  along the four directions are pictured as end-point and mid-point curves (so resulting as unidimensional  $IV$ -functions) red colored for canonical directions (*bottom-left*) and cyan for diagonals (*bottom-right*).

The  $i$ -th component of  $L(x)$  is then given by

$$L_i(x) = \left( \sum_{j=1}^n \hat{l}_{i,j} x_j; \left| \sum_{j=1}^n \tilde{w}_{i,j} x_j \right| \right) \text{ for all } i = 1, 2, \dots, m. \quad (65)$$

Consequently, the  $gH$ -differentiability (Fréchet, Gateaux, partial and directional  $gH$ -derivatives) is easily extended to vector-valued  $IV$  functions.

We conclude this section with the important example induced by a  $gH$ -linear vector-valued combination of intervals.

**Example 63.** Consider two real matrices  $\hat{A} = [\hat{a}_{i,j}]$ ,  $\tilde{A} = [\tilde{a}_{i,j}]$  of order  $(m, n)$  and two (column) vectors  $\hat{b}, \tilde{b} \in \mathbb{R}^m$ ; we obtain a vector  $IV$ -function  $F : \mathbb{R}^n \rightarrow I(\mathbb{R}^m)$ ,  $F = (\hat{F}; \tilde{F})$  by setting  $\hat{F}(x) = \hat{A}x + \hat{b}$  and  $\tilde{F}(x) = |\tilde{A}x + \tilde{b}|$  (here, the absolute value of a vector is obtained by the absolute values of each component).

The  $m$  components of  $F$  are the affine  $IV$ -functions

$$F_i(x) = \left( \hat{b}_i + \sum_{j=1}^n \hat{a}_{i,j} x_j; \left| \tilde{b}_i + \sum_{j=1}^n \tilde{a}_{i,j} x_j \right| \right), i = 1, 2, \dots, m; \quad (66)$$

each of them is Fréchet  $gH$ -differentiable and the  $gH$ -differential of  $F$  at  $x \in \mathbb{R}^n$  with respect to  $h \in \mathbb{R}^n$ , is the  $gH$ -Jacobian  $J_{gH}F(x)(h) = (\hat{A}h; |\tilde{A}h|)$ .

## 5.2. Higher order Fréchet differentiability and the $gH$ -Hessian matrix

Consider first the case of a  $IV$ -function of a single variable  $F : ]a, b[ \subset \mathbb{R} \rightarrow I(\mathbb{R})$ ,  $F(x) = (\hat{f}(x); \tilde{f}(x))$  with first order  $gH$ -derivative  $F'_{gH}(x) = \left( \frac{d}{dx} \hat{f}(x); |\tilde{w}_F(x)| \right)$  at all points in a neighborhood of a point  $x^{(0)} \in ]a, b[$ ; we know that  $\tilde{w}_F(x) \in \left\{ \frac{d}{dx^-} \tilde{f}(x), \frac{d}{dx^+} \tilde{f}(x) \right\}$  and  $F'_{gH}(x)$  exists if and only if  $\frac{d}{dx} \hat{f}(x)$  exists and the left and right derivatives  $\frac{d}{dx^-} \tilde{f}(x)$ ,  $\frac{d}{dx^+} \tilde{f}(x)$  have the same absolute value.

The second order  $gH$ -derivative of  $F$  at  $x^{(0)}$  is then

$$F''_{gH}(x^{(0)}) := \frac{d_{gH}}{dx} F'(x^{(0)}) = \lim_{h \rightarrow 0} \frac{F'_{gH}(x^{(0)} + h) \ominus_{gH} F'_{gH}(x^{(0)})}{h}. \quad (67)$$

On the other hand, we know that the limit (67) exists if and only if the second derivative of  $\hat{f}$  exists at  $x^{(0)}$  and the left and right derivatives of  $\tilde{w}_F(x)$  exist at  $x^{(0)}$  and have the same absolute value, i.e., for  $x = x^{(0)}$ ,

$$F''_{gH}(x) = \left( \frac{d^2}{dx^2} \hat{f}(x); |\tilde{w}_{F'}(x)| \right). \quad (68)$$

By continuing inductively, the  $gH$ -derivative of order  $r + 1$ ,  $r \geq 1$ , is given in terms of the  $r$ -order  $gH$ -derivative  $F^{(r)}_{gH}(x) = \left( \frac{d^r}{dx^r} \hat{f}(x); |\tilde{w}_{F^{(r-1)}}(x)| \right)$  of  $F$  (here,  $F^{(0)} = F$ ), i.e., of the  $(r + 1)$ -th derivative of  $\hat{f}$  and the left/right derivatives of  $\tilde{w}_{F^{(r-1)}}(x)$ , if they have the same absolute value:

$$F^{(r+1)}_{gH}(x) = \left( \frac{d^{(r+1)}}{dx^{(r+1)}} \hat{f}(x); |\tilde{w}_{F^{(r)}}(x)| \right).$$

To define Fréchet  $gH$ -differentiability of order  $r > 1$  for a  $IV$ -function of multiple variables,  $F : K \subset \mathbb{R}^n \rightarrow I(\mathbb{R})$ ,  $F(x) = (\hat{f}(x); \tilde{f}(x))$ , we need first to define “ $gH$ -multilinear” functions (applications)  $L : (\mathbb{R}^n)^r \rightarrow I(\mathbb{R})$ .

In this paper, we consider only  $r = 2$  and symmetric  $gH$ -bilinear  $IV$ -functions. A real-valued (symmetric) bilinear function on  $\mathbb{R}^n$  is any  $l : (\mathbb{R}^n)^2 \rightarrow \mathbb{R}$ ,  $(h, k) \rightarrow l(h, k) \in \mathbb{R}$ , such that for all  $h, k, h^{(1)}, h^{(2)}, k^{(1)}, k^{(2)} \in \mathbb{R}^n$  and all  $\alpha_1, \alpha_2 \in \mathbb{R}$ ,

$$l(\alpha_1 h^{(1)} + \alpha_2 h^{(2)}, k) = \alpha_1 l(h^{(1)}, k) + \alpha_2 l(h^{(2)}, k),$$

$$l(h, \alpha_1 k^{(1)} + \alpha_2 k^{(2)}) = \alpha_1 l(h, k^{(1)}) + \alpha_2 l(h, k^{(2)}),$$

$$l(h, k) = l(k, h) \text{ and, consequently,}$$

$$l(e_i, e_j) = l_{i,j} \in \mathbb{R}, \text{ for all canonical } e_i, e_j \in \mathbb{R}^n.$$

**Definition 64.** A  $gH$ -bilinear (symmetric)  $IV$ -function  $L : (\mathbb{R}^n)^2 \rightarrow I(\mathbb{R})$  has the form

$$L(h, k) = \left( \hat{l}(h, k); |\tilde{w}(h, k)| \right)$$

where  $\hat{l}(h, k)$ ,  $\tilde{w}(h, k)$  are two real-valued (symmetric) bilinear functions on  $\mathbb{R}^n$ .

The second order Fréchet  $gH$ -differentiability is defined as follows.

**Definition 65.** Let  $F : K \subseteq \mathbb{R}^n \rightarrow I(\mathbb{R})$  be an interval-valued function with (first order)  $gH$ -differential function  $D_{gH} F(x)$  such that  $D_{gH} F(x)(h) = L_x^{(1)}(h)$  and  $L_x^{(1)}(h) = \left( h^T \hat{l}_x^{(1)}; h^T \tilde{w}_x^{(1)} \right)$ ,  $h \in \mathbb{R}^n$ , at all points  $x$  in a neighborhood of an internal point  $x^{(0)} \in K$ .

We say that  $F$  is twice Fréchet  $gH$ -differentiable at  $x^{(0)}$  if there exists a  $gH$ -bilinear (symmetric)  $IV$ -function  $L_{x^{(0)}}^{(2)} : (\mathbb{R}^n)^2 \rightarrow I(\mathbb{R})$ , depending on  $x^{(0)}$ , with  $L_{x^{(0)}}^{(2)}(h, k) = \left( \hat{l}^{(2)}(h, k); |\tilde{w}^{(2)}(h, k)| \right)$  such that

$$\lim_{\|k\| \rightarrow 0} \frac{\left\| \left( D_{gH} F(x^{(0)} + k)(h) \ominus_{gH} D_{gH} F(x^{(0)})(h) \right) \ominus_{gH} L_{x^{(0)}}^{(2)}(h, k) \right\|_H}{\|k\|} = 0 \quad (69)$$

uniformly for  $h$  in a bounded subset of  $\mathbb{R}^n$ . The interval-valued function  $L_{x^{(0)}}^{(2)} : (h, k) \rightarrow L_{x^{(0)}}^{(2)}(h, k)$ ,  $h, k \in \mathbb{R}^n$ , is called the second order Fréchet  $gH$ -derivative (or second order Fréchet  $gH$ -differential function) of  $F$  at  $x^{(0)}$ , denoted as  $D_{gH}^{(2)} F(x^{(0)})$ , i.e.,  $D_{gH}^{(2)} F(x^{(0)})(h, k) = L_{x^{(0)}}^{(2)}(h, k)$  for all  $h, k \in \mathbb{R}^n$ .

It is easy to see that the  $gH$ -bilinear function  $L_{x^{(0)}}^{(2)}$  can be obtained as follows (we omit reference to  $x^{(0)}$  for simplicity):

**Proposition 66.** *There exist two symmetric square matrices of order  $n$ , say  $\hat{\mathbf{L}} = [\hat{l}_{i,j}]$ ,  $\tilde{\mathbf{L}} = [\tilde{l}_{i,j}]$  such that the IV-function  $(h, k) \rightarrow \mathbf{L}^{(2)}(h, k)$  satisfying (69) is obtained as*

$$\mathbf{L}^{(2)}(h, k) = \left( \sum_{i,j=1}^n \hat{l}_{i,j} h_i k_j; \left| \sum_{i,j=1}^n \tilde{l}_{i,j} h_i k_j \right| \right). \quad (70)$$

**Example 67.** Let  $n = 2$  and  $F = (\hat{f}; \tilde{f})$ ,  $x \in [-1, 1]^2$  given by

$$\begin{aligned} \hat{f}(x_1, x_2) &= x_1^2 + x_1 x_2 + x_2^2 - x_1 + 1 \text{ and} \\ \tilde{f}_w(x_1, x_2) &= x_1^2 - x_2^2 + x_1 - x_2 + 3, \tilde{f}(x_1, x_2) = \left| \tilde{f}_w(x_1, x_2) \right|. \end{aligned}$$

Consider  $x^{(0)} = (0, 0)$ ,  $F(x^{(0)}) = (1; 3) = [-2, 4]$ ,  $\tilde{f}_w(x^{(0)}) = 3$ .

The two functions  $\hat{f}, \tilde{f}_w$  are differentiable at all  $x \in \mathbb{R}^2$  with gradient functions at  $x$  given by  $\nabla \hat{f}(x) = (2x_1 + x_2 - 1, x_1 + 2x_2)^T$ ,  $\nabla \hat{f}(0) = (-1, 0)^T$  and  $\nabla \tilde{f}_w(x) = (2x_1 + 1, -2x_2 - 1)^T$ ,  $\nabla \tilde{f}_w(0) = (1, -1)^T$ . It follows that  $F(x_1, x_2)$  is Fréchet  $gH$ -differentiable.

The  $gH$ -differential function at the origin  $x^{(0)}$  is  $\mathbf{L}(h) = (-h_1; |x_1 - x_2|)$  with  $\hat{l} = (-1, 0)^T$ ,  $\tilde{w} = (1, -1)^T$ , then the interval-valued first-order  $gH$ -tangency function is  $\mathbf{P}_{x^{(0)}}^* = (\hat{p}^*; |\tilde{p}_w^*|)$  with  $\hat{p}^*(x) = 1 - x_1$  and  $\tilde{p}_w^*(x) = 3 + x_1 - x_2$ .

The interval  $gH$ -Hessian matrix of  $F(x)$  at  $x^{(0)}$  is  $\mathbf{H} = (\hat{H}; |\tilde{H}_w|)$  where the (midpoint and radius component) two matrices are, respectively,

$$\hat{H} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \tilde{H}_w = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}.$$

For this simple case, where  $\hat{f}, \tilde{f}_w$  are differentiable, we can easily obtain the second order  $gH$ -tangency function to  $F$  at  $x^{(0)}$ , given by

$$\mathbf{Q}_{x^{(0)}}^*(x) = (\hat{Q}^*(x); |\tilde{Q}_w^*(x)|),$$

where, with  $\hat{p}^*(x), \tilde{p}_w^*(x)$  as above,

$$\begin{aligned} \hat{Q}^*(x) &= \hat{p}^*(x) + \frac{1}{2}(x - x^{(0)})^T \hat{H}(x - x^{(0)}), \\ \tilde{Q}_w^*(x) &= \tilde{p}_w^*(x) + \frac{1}{2}(x - x^{(0)})^T \tilde{H}_w(x - x^{(0)}). \end{aligned}$$

In Fig. 7 we picture function  $F(x)$  with both the first-order tangent IV-function  $\mathbf{P}_{x^{(0)}}^*(x)$  and the second-order tangent IV-function  $\mathbf{Q}_{x^{(0)}}^*(x)$ .

## 6. Concluding comments and further work

The definition of  $gH$ -linear interval-valued functions, introduced in this paper, as a tool to define Fréchet and Gateaux  $gH$ -differentiability, appears to be of great interest and will possibly offer the setting for further progress in the study of interval-valued functions in a very general series of applications.

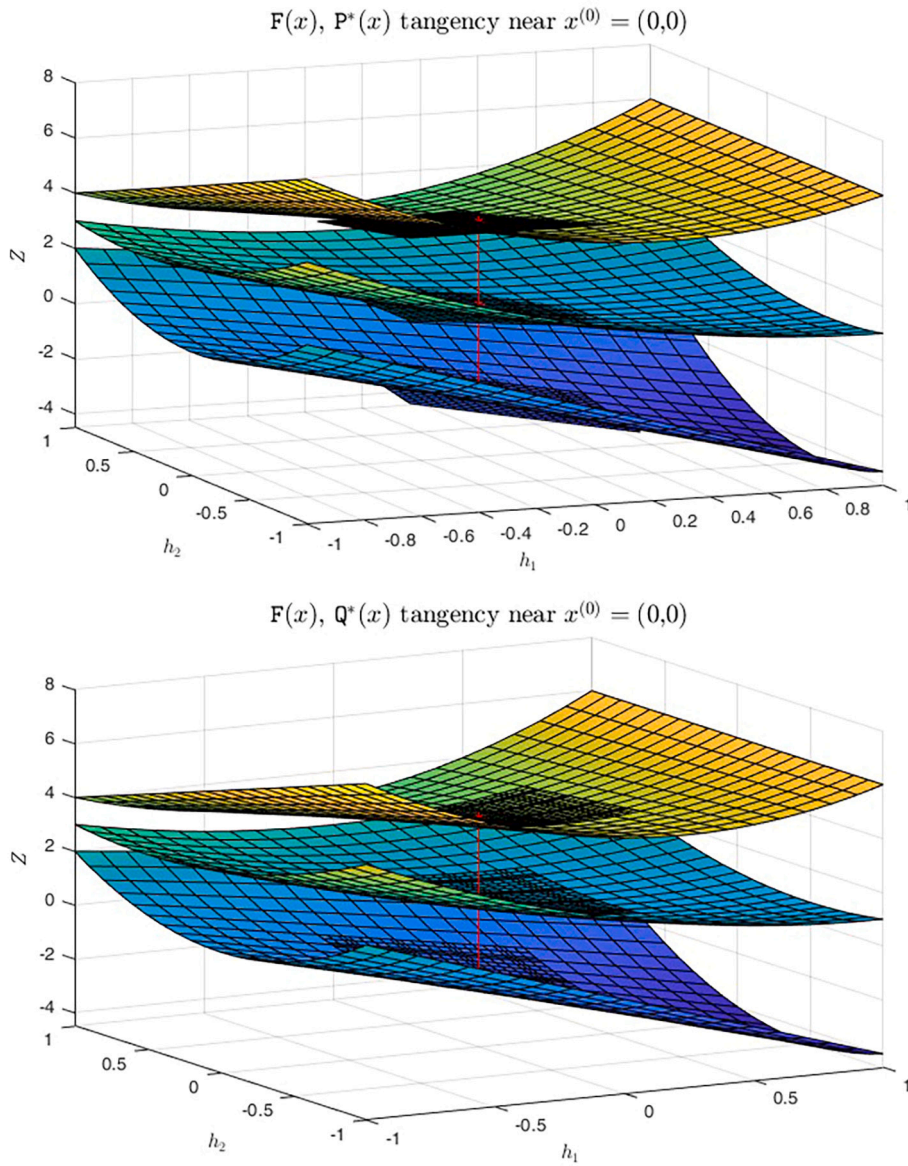
Our attempt to preserving an approach based on some form of linearity, allows a definition of  $gH$ -differential IV-function which substantially adheres to the classical theory of differentiable functions (with the fundamental properties of first-order tangency and linear-type approximation) and to its connections with directional and partial derivatives. Indeed, as we have seen in section 4, Minkowski-type combinations of partial derivatives will not be able to produce, in general, the directional  $gH$ -derivatives of an IV-function and a different combination is required, based on a  $gH$ -linear function.

Based on the results presented in sections 4 and 5, further work is planned to examine the possibilities offered by these new tools and to visit their potential in several areas of research. Some important applications are planned in interval-valued optimization, to get non-dominated solutions for constrained or unconstrained minimization or maximization problems with IV-functions of multiple variables (see [5], [6] for recent results).

A second important application is to characterize the  $gH$ -differential of a fuzzy valued function of several variables, as we outlined in our paper [45] and in some conference proceedings. In particular, the concept of level-wise fuzzy LgH-differentiability, first introduced in [9] for the single variable case and studied also by other authors, can immediately allow the extension of Fréchet  $gH$ -derivative for fuzzy-valued functions of multiple variables.

### 6.1. Generalization to fuzzy-valued functions

When considering fuzzy-valued functions level-wise, expressed in terms of their  $\alpha$ -cuts, the concepts of fuzzy  $gH$ -difference and Fréchet  $gH$ -differentiability are similar to the ones for vector-valued IV-functions but with an infinite number of components, corresponding to  $\alpha \in [0, 1]$ .



**Fig. 7.** This figure refers to the  $IV$ -function  $F(x)$  described in Example 67, with  $x \in [-1, 1]^2$ . The first order tangent function  $P^*(x)$  (Top picture) and the second order  $gH$ -tangent function  $Q^*(x)$  (Bottom picture) are computed with  $x$  in a neighborhood of the origin.

The extension of Fréchet/Gateaux  $gH$ -differentiability to fuzzy-valued functions seems to be almost immediate, in particular if one considers the concept of level-wide  $gH$ -differentiability (LgH-differentiability for short).

We say that a fuzzy-valued function  $F : X \subseteq \mathbb{R}^n \longrightarrow \mathbb{R}_{\mathcal{F}}$  having  $\alpha$ -cuts  $[F(x)]_{\alpha} = (\hat{F}_{\alpha}(x); \tilde{F}_{\alpha}(x))$  is Gateaux (or Fréchet) LgH-differentiable, if and only if all interval-valued functions  $x \longrightarrow (\hat{F}_{\alpha}(x); \tilde{F}_{\alpha}(x))$ , with a uniformity condition for  $\alpha \in [0, 1]$ , are  $gH$ -differentiable.

A LgH-linear function is such that  $[L(x)]_{\alpha} = (\hat{L}_{\alpha}(x); \tilde{L}_{\alpha}(x))$  with linear functions  $\hat{L}_{\alpha}(x)$  and  $\tilde{L}_{\alpha}(x)$ . Clearly, all the results of previous sections are applied to the level-wise fuzzy case with the additional requirement that they are valid for all  $\alpha \in [0, 1]$ .

On the other hand, the concepts and results on fuzzy total and directional LgH-derivatives, as obtained in [45], can be immediately applied to characterize Gateaux and Fréchet LgH-differentiability.

Concepts of Gateaux and Fréchet differentiability of fuzzy functions have been recently addressed, e.g., in [16], [38], in the setting of so called  $A$ -linearly correlated (interval or) fuzzy functions (here,  $A \in \mathbb{R}_{\mathcal{F}}$ ).

In particular, the case of fuzzy symmetric  $A$  has been deeply examined in [39] and it will be interesting to attempt a comparison of the two approaches; to this aim, it will be useful to consider the so-called crisp-profile decomposition of  $F(x) \in \mathbb{R}_{\mathcal{F}}$  (described, e.g., in [44]): by writing its  $\alpha$ -cuts in midpoint notation

$$[F(x)]_\alpha = \left( \hat{F}_\alpha(x); \tilde{F}_\alpha(x) \right) = \hat{F}_\alpha(x) + [-\tilde{F}_\alpha(x), \tilde{F}_\alpha(x)],$$

we have that, for each  $\alpha \in [0, 1]$ ,  $[F(x)]_\alpha$  is the sum of an ordinary function  $\tilde{F}_\alpha(x)$  (called the profile of  $F(x)$ ) and a 0-symmetric interval function  $[-\tilde{F}_\alpha(x), \tilde{F}_\alpha(x)]$ , similar to the construction in [39].

## 6.2. Further work on applications

In this paper, the case of vector interval valued functions is only outlined, and much more can be done in view of numerous applications, e.g., in multi-objective fuzzy optimization. The same can be said for higher-order gH-differentiable functions and in particular on the gH-Hessian matrix, with possible characterizations of local/global concavity or convexity and conditions for minimum or maximum points (interval and fuzzy cases); remark, e.g., that the gH-tangent functions  $P^*(x)$  and  $Q^*(x)$  (not simply the gH-gradient), given in this paper, require the Fréchet gH-differential to be correctly applied. It will be important to examine the gH-Hessian matrix, in connection with concavity and convexity properties and with conditions for LU-order maximum and minimum points, among other topics.

Under the framework of the present paper, a proposal of descent directions using gH-differentiability and gH-derivatives will be object of study, as well as a related algorithm to obtain non-dominated solutions to IV-minimization problems. In particular, for a function  $F(x) = (\hat{f}(x); \tilde{f}(x))$ , the existence of descent directions for the ordinary function  $\hat{f}(x)$  (the midpoint component of  $F(x)$ ) is sufficient to the existence of decreasing directions for  $F(x)$  (with respect to the partial LU-order); furthermore, similarly to the single-variable case examined in [48], the characterization and search for non-dominated solutions and the identification of the (local) efficient frontier (for minima or maxima) can be analyzed efficiently using the midpoint representation of  $F(x)$  and the corresponding expressions of the gH-derivatives.

## CRediT authorship contribution statement

**Luciano Stefanini:** Writing – original draft, Validation, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Manuel Arana-Jiménez:** Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Laerte Sorini:** Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix A. Notation for intervals used in the paper

We need a remark on possible notation for intervals, to choose among the many existing ones and to allow for the two mostly adopted end-point and mid-point representations; we have tried a compromise among existing practice and preferences from hundreds of publications.

1)  $\mathbb{R}, \mathbb{R}^n, \mathbb{R}^{(m,n)}$  denote the spaces of reals, (column)  $n$ -vectors of reals,  $(m, n)$ -matrices of reals, respectively:  $a, b, c, x, a_i, b_i, a_{i,j}, \alpha, \lambda, \mu$  generic reals;  $A, B, A_i, B_j$  generic sets;  $x = (x_1, \dots, x_n)^T$  (column)  $n$ -vector of reals;  $\mathbf{A} = [a_{i,j}]$ ,  $\underline{\mathbf{A}} = [\underline{a}_{i,j}] \in \mathbb{R}^{(m,n)}$   $(m, n)$ -matrices of reals.

2)  $A, B, C, X, A_1, B_j$  denote intervals:  $A = [\underline{a}, \bar{a}]$  interval in *end-point representation*  $\underline{a} \leq \bar{a}$ ; similarly  $X = [\underline{x}, \bar{x}]$  with  $\underline{x} \leq \bar{x}$ , etc. For an interval  $X = [\underline{x}, \bar{x}]$ , we denote the midpoint  $\hat{x} = \frac{1}{2}(\underline{x} + \bar{x})$  and the radius  $\tilde{x} = \frac{1}{2}(\bar{x} - \underline{x}) \geq 0$  so that, equivalently,  $X = [\hat{x} - \tilde{x}, \hat{x} + \tilde{x}]$ ;  $A = (\hat{a}; \tilde{a})$ ,  $\tilde{a} \geq 0$  or  $X = (\hat{x}; \tilde{x})$ ,  $\tilde{x} \geq 0$  is the *mid-point representation* of intervals  $A$  or  $X$ , respectively.

3)  $X = (X_1, \dots, X_n)^T$  denotes a column  $n$ -vector of intervals ( $n$ -box),  $X_i = [\underline{x}_i, \bar{x}_i] = (\hat{x}_i; \tilde{x}_i)$ ,  $\underline{x}_i \leq \bar{x}_i$ ,  $\tilde{x}_i \geq 0$ . If  $\underline{x} = (\underline{x}_1, \dots, \underline{x}_n)^T$ ,  $\bar{x} = (\bar{x}_1, \dots, \bar{x}_n)^T$  are real  $n$ -vectors,  $X = [\underline{x}, \bar{x}]$  is an  $n$ -vector of intervals with  $\underline{x} \leq \bar{x}$  component-wise. If  $\hat{x} = (\hat{x}_1, \dots, \hat{x}_n)^T$ ,  $\tilde{x} = (\tilde{x}_1, \dots, \tilde{x}_n)^T$  are real  $n$ -vectors, with  $\tilde{x} \geq 0$  we denote  $X = (\hat{x}; \tilde{x})$ .  $[A_{i,j}]$  denotes a  $(m, n)$ -matrix of intervals  $A_{i,j} = [\underline{a}_{i,j}, \bar{a}_{i,j}] = (\hat{a}_{i,j}; \tilde{a}_{i,j})$ ,  $\underline{a}_{i,j} \leq \bar{a}_{i,j}$ ,  $\tilde{a}_{i,j} \geq 0$ ,  $i = 1, 2, \dots, m$ ,  $j = 1, 2, \dots, n$ . If  $\underline{\mathbf{A}} = [\underline{a}_{i,j}]$ ,  $\bar{\mathbf{A}} = [\bar{a}_{i,j}] \in \mathbb{R}^{(m,n)}$  are real  $(m, n)$ -matrices we denote  $[A_{i,j}] = [\underline{\mathbf{A}}, \bar{\mathbf{A}}]$  a  $(m, n)$ -matrix on intervals, with matrices  $\underline{\mathbf{A}} \leq \bar{\mathbf{A}}$  element by element. Correspondingly,  $\hat{\mathbf{A}} = [\hat{a}_{i,j}] \in \mathbb{R}^{(m,n)}$ ,  $\tilde{\mathbf{A}} = [\tilde{a}_{i,j}] \in \mathbb{R}^{(m,n)}$  are matrices with midpoint and radius values, so that also  $[A_{i,j}] = (\hat{\mathbf{A}}; \tilde{\mathbf{A}})$  with  $\tilde{\mathbf{A}} \geq 0$  element by element and  $\underline{\mathbf{A}} = \hat{\mathbf{A}} - \tilde{\mathbf{A}}$ ,  $\bar{\mathbf{A}} = \hat{\mathbf{A}} + \tilde{\mathbf{A}}$ .

4)  $I(\mathbb{R})$  denotes the set of real intervals;  $I(\mathbb{R})^n$  the set of all  $n$ -vectors of intervals ( $n$ -boxes);  $I(\mathbb{R})^{(m,n)}$  set of all  $(m, n)$ -matrices of intervals.

## Data availability

No data was used for the research described in the article.

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